CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

2.1 Equations of Motion

Syllabus Section: 2. Kinematics **Paper Type:** Structured Questions (Paper 2) **Total:** 13 Questions (157 Marks)

Question 1 Easy [16 marks]

Part (a) [4 marks]

State the difference between

(i) distance and displacement,

[2]

(ii) speed and velocity.

[2]

Mark Scheme (4 marks):

(i) The difference between distance and displacement is:

- Distance is defined as the total length between two points regardless of direction **OR** distance is a scalar quantity; *[1 mark]*
- Displacement is defined as the net change in an object's position **OR** displacement is a vector quantity; *[1 mark]*

(ii) The difference between speed and velocity is:

- Speed is defined as the total distance travelled per unit of time **OR** speed is a scalar quantity; *[1 mark]*
- Velocity is defined as the rate of change of displacement of an object **OR** velocity is a vector quantity; *[1 mark]*

[Total: 4 marks]

Part (b) [6 marks]

Complete the table in Table 1.1 to describe the features of displacement-time, velocity-time and acceleration-time graphs represent.

Place a cross (✗) in any grid space which does not represent anything. Some grid spaces have been completed for you.

Table 1.1

	displacement-time	velocity-time	acceleration-time
gradient represents			✗
area under curve represents			

y-intercept represents			
horizontal line represents	zero velocity		
straight section represents			
curved section represents		non-uniform acceleration	

Mark Scheme (6 marks):

The correctly complected table is as follows:

	displacement-time	**velocity-time**	**acceleration-time**
gradient represents	***velocity***	***acceleration***	✗
area under curve represents	✗	***displacement***	***change in velocity***
y-intercept represents	***initial position***	***initial velocity***	***initial acceleration***
horizontal line represents	zero velocity	***constant velocity***	***constant / uniform acceleration***
straight section represents	***constant velocity***	***constant / uniform acceleration***	✗
curved section represents	***changing velocity / acceleration***	non-uniform acceleration	✗

- *One mark for each correct row*

[Total: 6 marks]

Part (c) [3 marks]

On Fig. 1.2, sketch the variation of time with distance, velocity, and acceleration for an object moving with **constant velocity**.

Take the initial position to be zero.

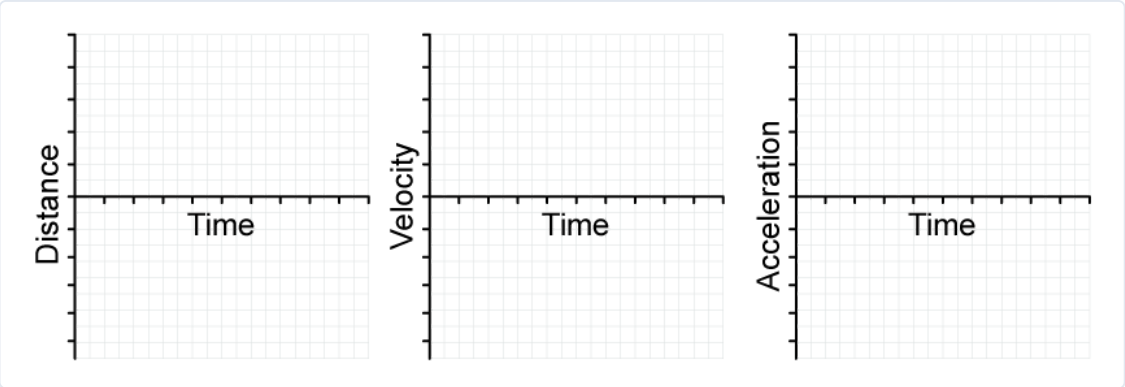
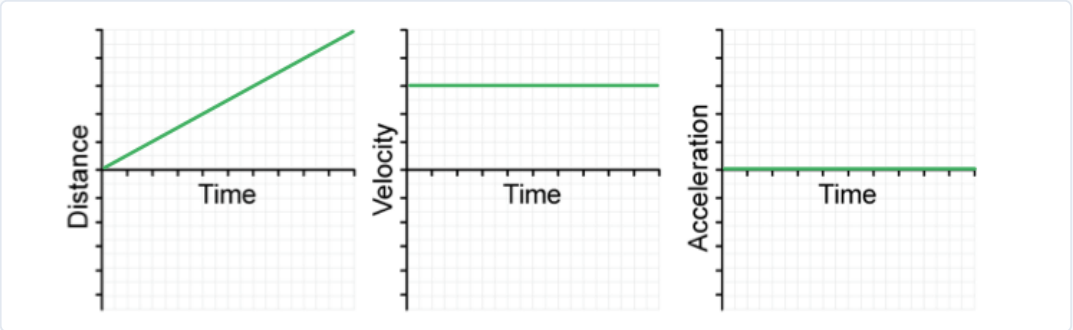


Fig. 1.2

Mark Scheme (3 marks):

For an object moving with constant velocity:



- Distance-time graph: straight, diagonal line starting from zero drawn; *[1 mark]*
- Velocity-time graph: straight, horizontal non-zero line drawn; *[1 mark]*
- Acceleration-time graph: straight, horizontal drawn at zero; *[1 mark]*

[Total: 3 marks]

Part (d) [3 marks]

On Fig. 1.3, sketch the variation of time with distance, velocity, and acceleration for an object moving with **uniform acceleration**.

Take the initial position to be zero.

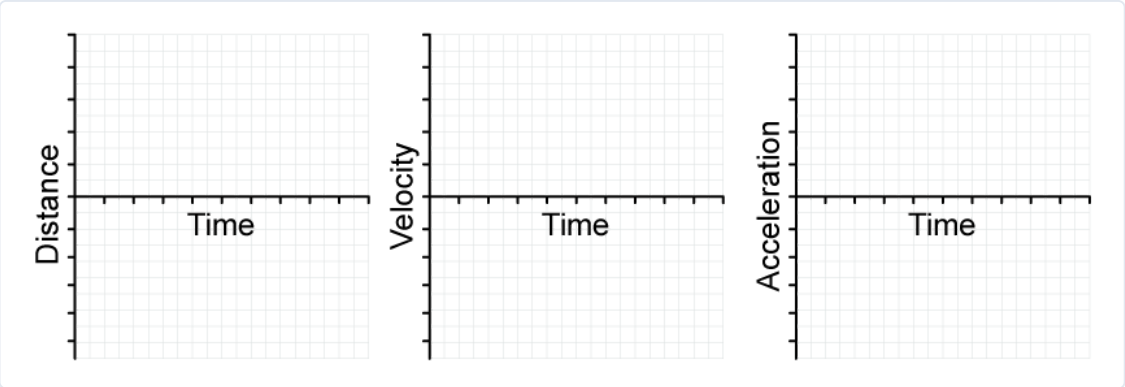
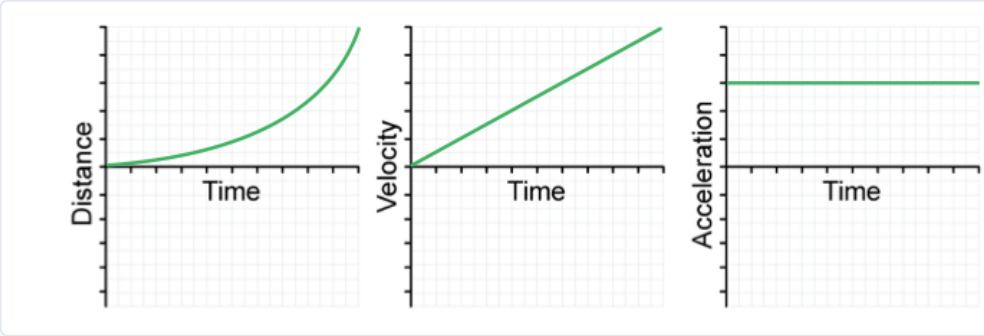


Fig. 1.3

Mark Scheme (3 marks):

For an object moving with uniform acceleration:



- Distance-time graph: curved / parabolic line starting from zero drawn; *[1 mark]*
- Velocity-time graph: straight, diagonal line starting from zero drawn; *[1 mark]*
- Acceleration-time graph: straight, horizontal non-zero line drawn; *[1 mark]*

[Total: 3 marks]

Question 2 Easy [12 marks]

Part (a) [2 marks]

Place ticks (✓) in the table in Table 1.1 to indicate which of the quantities are vectors and which are scalars.

Table 1.1

	velocity	speed	distance	displacement	acceleration
vector					
scalar					

Mark Scheme (2 marks):

A correctly completed table is as follows:

	velocity	speed	distance	displacement	acceleration
vector	✓			✓	✓
scalar		✓	✓		

- Vectors: velocity, displacement, acceleration; *[1 mark]*
- Scalar: speed, distance; *[1 mark]*

[Total: 2 marks]

Part (b) [3 marks]

Two of the equations of uniformly accelerated motion are

$$s = ut + \tfrac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

(i) the definition of uniform acceleration,

[1]

(ii) the **two** other equations of motion that can be used to derive the above equations.

[2]

Mark Scheme (3 marks):

(i) Uniform acceleration is defined as:

*Any **one** from:*

- When an object's velocity increases (or decreases) at a constant rate; *[1 mark]*
- Acceleration in which the rate of change of velocity is constant; *[1 mark]*
- When the velocity of an object changes by equal amounts in equal time periods; *[1 mark]*

(ii) The two other equations of motion are:

- $v = u + at$ *[1 mark]*
- $s = \frac{u + v}{2}t$ *[1 mark]*

[Total: 3 marks]

Part (c) **[5 marks]**

Fig. 1.2 shows an electric motor which is used to lift and lower heavy loads.

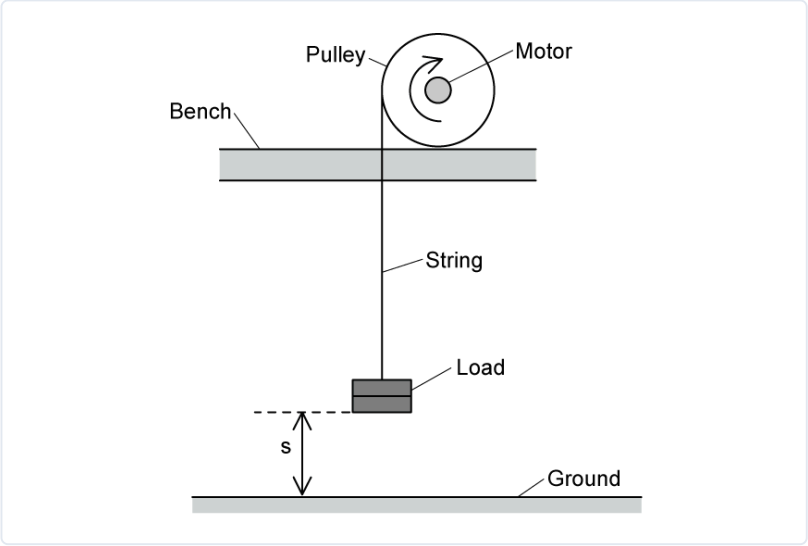


Fig. 1.2

Initially, the load is on the ground. Fig. 1.3 shows the variation of the displacement *s* of the load with time *t*.

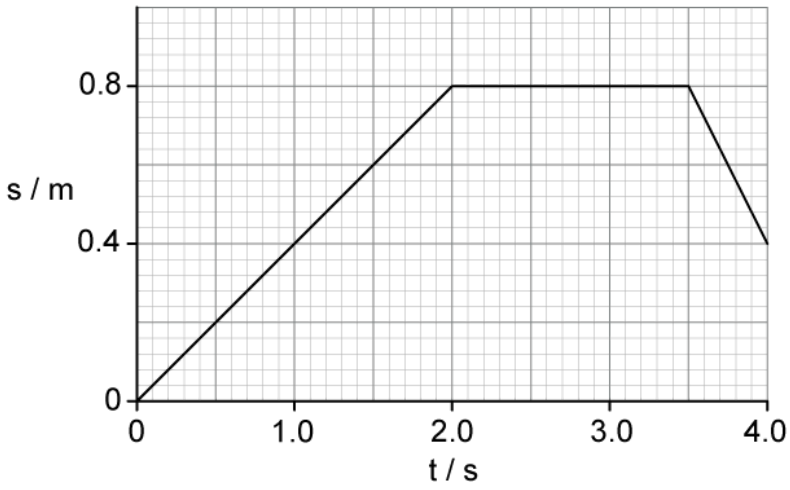


Fig. 1.3

(i) Complete Table 1.4 to show the magnitude and direction of the velocity of the load during the time intervals shown.

Table 1.4

time interval	magnitude of velocity	direction of velocity
t = 0 to 2.0 s		
t = 2.0 to 3.5 s		
t = 3.5 to 4.0 s		

[3]

(ii) On Fig. 1.5, sketch a graph to show the variation of the velocity v of the load with time t.

Include values for velocity on the v-axis.

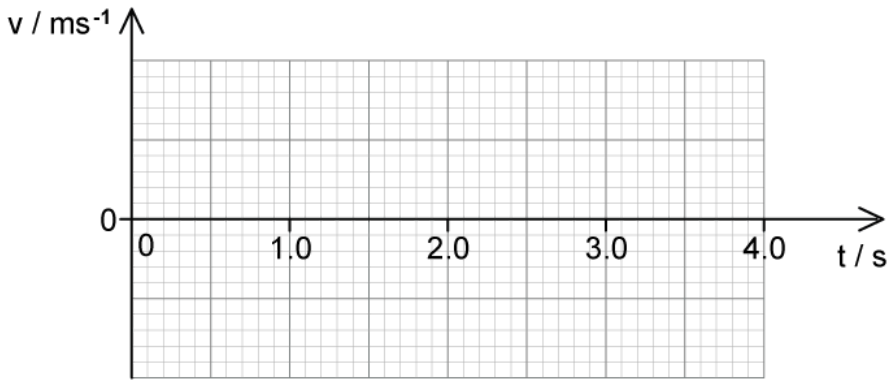


Fig. 1.5

[2]

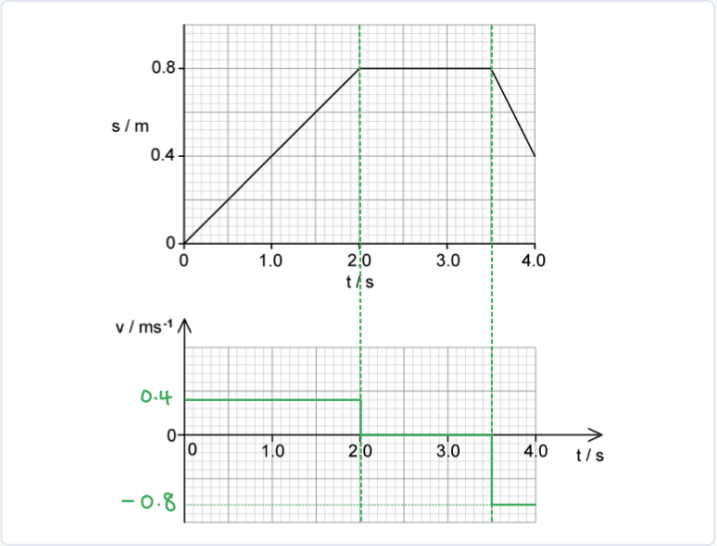
Mark Scheme (5 marks):

(i) The variation of the velocity of the load from t = 0 to 4.0 s is:

time interval	magnitude of velocity	direction of velocity
t = 0 to 2.0 s	***0.4 m s^{-1}***	***upwards / positive***
t = 2.0 to 3.5 s	***0***	***none***
t = 3.5 to 4.0 s	***0.8 m s^{-1}***	***downwards / negative***

- *One mark for each correct row*

(ii) The velocity-time graph is as follows:



- All lines drawn straight and horizontal; *[1 mark]*
- Values +0.4 m s^{-1} and −0.8 m s^{-1} in correct places on the *v*-axis; *[1 mark]*

Example of calculation:

- Upwards velocity: $v = \frac{\Delta s}{\Delta t} = \frac{0.8}{2.0} = 0.4 \ m \ s^{-1}$
- Downwards velocity: $v = \frac{\Delta s}{\Delta t} = \frac{0.4 - 0.8}{4.0 - 3.5} = -0.8 \ m \ s^{-1}$

[Total: 5 marks]

Part (d) [2 marks]

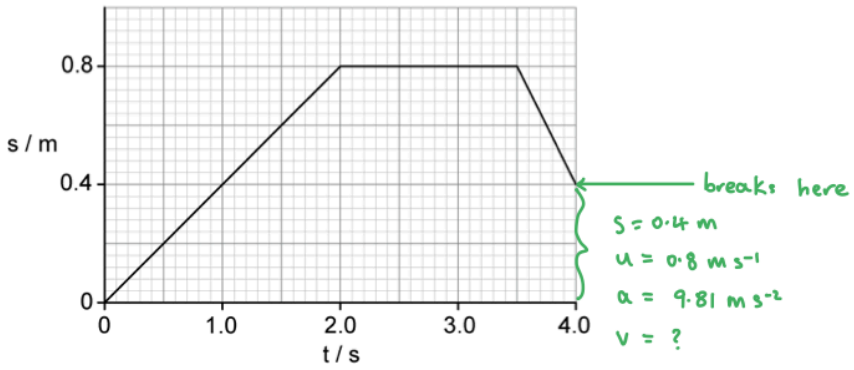
As the load is lowered towards the ground, the string breaks when *t* = 4.0 s. It then falls vertically toward the ground.

Calculate the velocity of the load just before it hits the ground.

Assume air resistance is negligible.

Mark Scheme (2 marks):

Calculate the velocity of the load just before it hits the ground:



List the known quantities, now defining downwards as the positive direction (*u* is now positive):

- Initial velocity, u = 0.80 m s^{-1} (from **(c)**)
- Acceleration, $a = g$ = 9.81 m s^{-2}
- Distance, s = 0.40 m
- Final velocity = v

Select the correct SUVAT equation:

- $v^2 = u^2 + 2as$

Substitute the values and evaluate:

- v^2 = 0.80^2 + (2 × 9.81 × 0.40) *[1 mark]*
- $v = \sqrt{8.488}$
- v = 2.913 = 2.9 m s^{-1} (2 s.f.) *[1 mark]*

[Total: 2 marks]

*Make sure to use the correct initial velocity here - the 0.40 m s^{-1} was the velocity as the load was being lifted **up** (not down, as this question is asking about.)*

*For questions like this, you can choose which direction is the positive direction, just be **clear** which you are doing in your calculations and don't mix your directions up. In this question, if we were going to keep **up** as the positive direction, our values would be:*

- *Initial velocity, $u = -0.80$ m s^{-1}*
- *Acceleration, $a = g = -9.81$ m s^{-2}*
- *Distance, $s = -0.40$ m*

Distance is covered downwards, acceleration acts downwards and initial velocity is downwards - all in the negative direction. It can be easier to just say "downwards is the positive direction" and avoid all those minus signs.

Question 3 Easy [9 marks]

Part (a) [3 marks]

A student drops a stone into a well and measures the time it takes to hear the stone splash at the bottom.

Explain how this measurement of time can be used to find the depth of the well.

Mark Scheme (3 marks):

This measurement of time can be used to find the depth of the well because:

- The initial velocity *u* is zero **AND** the acceleration (due to gravity) is g = 9.81 m s^{-2} *[1 mark]*
- Final velocity *v* can be found from $v = u + at$ **OR** $a = \frac{\Delta v}{t}$ *[1 mark]*
- The depth *s* can be found using $v^2 = u^2 + 2as$ and $s = ut + \frac{1}{2}at^2$ **OR** from the area under a graph of *v* against *t*; *[1 mark]*

[Total: 3 marks]

Part (b) [2 marks]

It takes 3.2 s for the stone to drop from rest and splash into the water at the bottom.

Calculate the velocity of the stone when it hits the water.

Mark Scheme (2 marks):

To calculate the velocity of the stone when it hits the water:

List the known quantities:

- Time, t = 3.2 s
- Acceleration, g = 9.81 m s^{-1}
- Initial velocity, u = 0

Calculate the final velocity *v*:

- $v = u + at$
- v = 0 + (9.81 × 3.2) *[1 mark]*
- v = 31.392 = 31 m s^{-1} (downward) *[1 mark]*

[Total: 2 marks]

Part (c) [2 marks]

On Fig. 1.1, sketch the variation of time with velocity for the stone as it falls down the well.

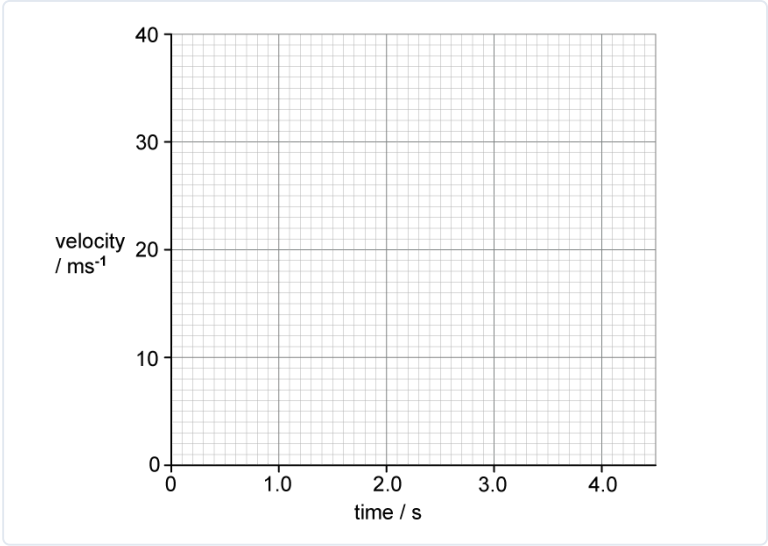


Fig. 1.1

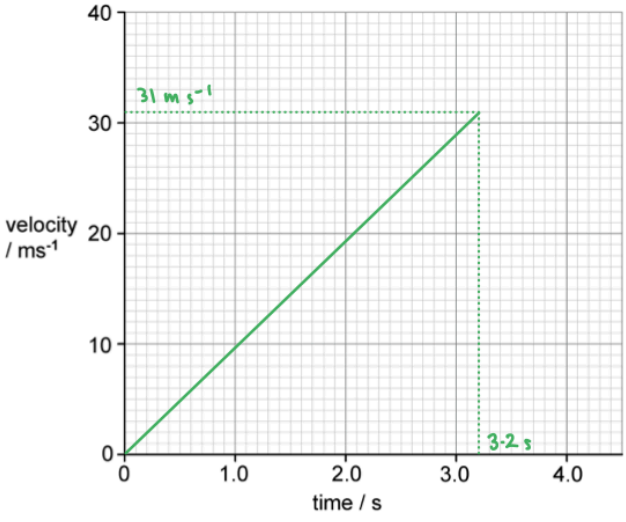
Mark Scheme (2 marks):

On Fig. 1.1, sketch the variation of time with velocity for the stone as it falls down the well.

List the known quantities:

- Final speed, v = 31 m s^{-1} (from part **(b)**)
- Time taken, t = 3.2 s

The variation of time with velocity for the stone is as follows:



- Straight line drawn starting from zero; *[1 mark]*
- Line stops at t = 3.2 s, v = 31 m s^{-1}; *[1 mark]*

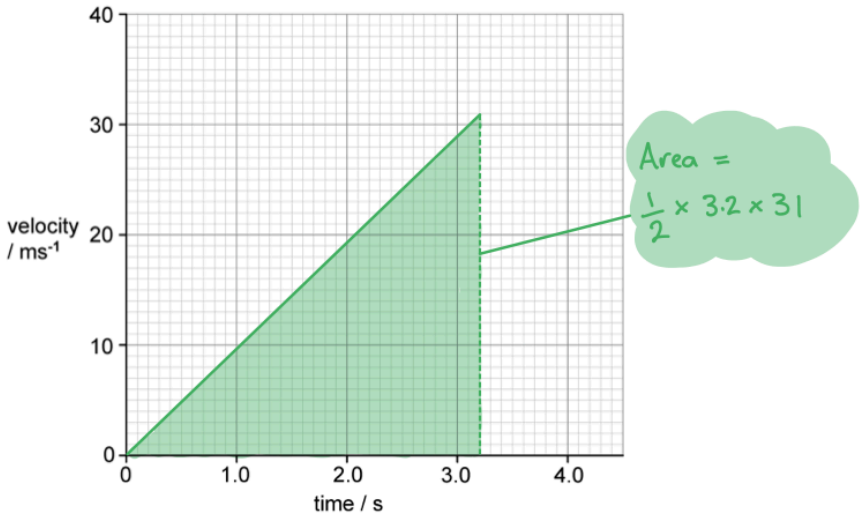
[Total: 2 marks]

Part (d) [2 marks]

Use your graph from **(c)** to determine the depth of the well.

Mark Scheme (2 marks):

Determine the depth of the well from the graph:



- Displacement = area under the velocity-time graph
- Depth = $\frac{1}{2} \times 3.2 \times 31$ *[1 mark]*
- Depth = 49.6 = 50 m *[1 mark]*

[Total: 2 marks]

Question 4 Easy [3 marks]

Part (a) [1 mark]

Define *velocity*.

Mark Scheme (1 marks):

Velocity is defined as:

- The rate of change of displacement

  OR

  The change in displacement per unit time *[1 mark]*

Part (b) [2 marks]

Fig. 1.1 shows the velocity–time graph for the car as it travels along a straight road.

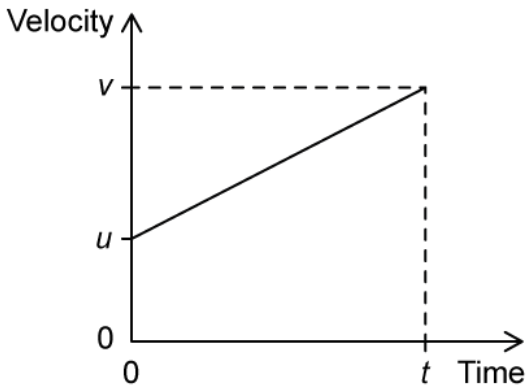


Fig. 1.1

(i) Determine an expression, in terms of u, v and t, for the area under the graph.

[1]

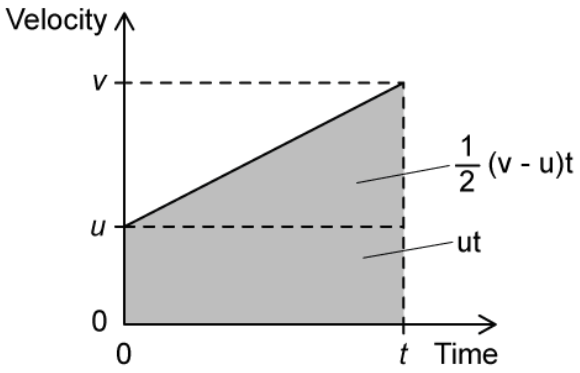
(ii) State the physical quantity represented by the area under the graph.

[1]

Mark Scheme (2 marks):

(i) To determine an expression, in terms of u, v and t, for the area under the graph:

- Identify the areas under the graph:



- Write an expression for the area under the graph:

area = ut + $\frac{1}{2}v$ - ut *[1 mark]*

Any **one** from:

- area = ut + $\frac{1}{2}v$ - ut
- area = vt - $\frac{1}{2}v$ - ut
- area = $\frac{1}{2}u$ + vt

[1 mark]

(ii) The physical quantity represented by the area under the graph is:

- Displacement *[1 mark]*

Question 5 Medium [13 marks]

Part (a) [1 mark]

Define acceleration.

Mark Scheme (1 marks):

Acceleration is defined as:

- The rate of change of velocity

  OR

- The change in velocity ÷ time taken; *[1 mark]*

[Total: 1 mark]

Part (b) [7 marks]

A ball is kicked from horizontal ground towards a vertical wall, as shown in Fig. 1.1.

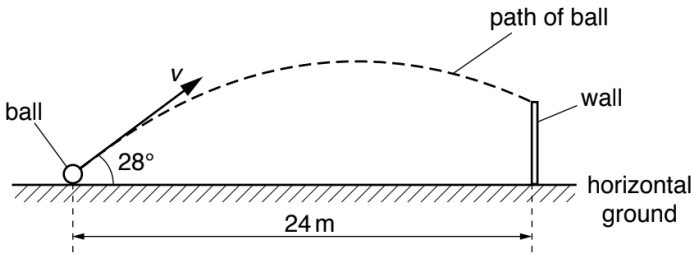


Fig. 1.1 (not to scale)

The horizontal distance between the initial position of the ball and the base of the wall is 24 m. The ball is kicked with an initial velocity v at an angle of 28° to the horizontal. The ball hits the top of the wall after a time of 1.5 s. Air resistance is negligible.

(i) Calculate the initial horizontal component v_x of the velocity of the ball.

v_x = ... m s^{-1} [1]

(ii) Show that the initial vertical component v_y of the velocity of the ball is 8.5 m s^{-1}.

[2]

(iii) Calculate the time taken for the ball to reach its maximum height above the ground.

time = ... s [2]

(iv) The ball is kicked at time t = 0. Assume that the vertical component v_y of the velocity of the ball is positive in the upwards direction.

On Fig. 1.2, sketch the variation with time t of v_y for the time until the ball hits the wall.

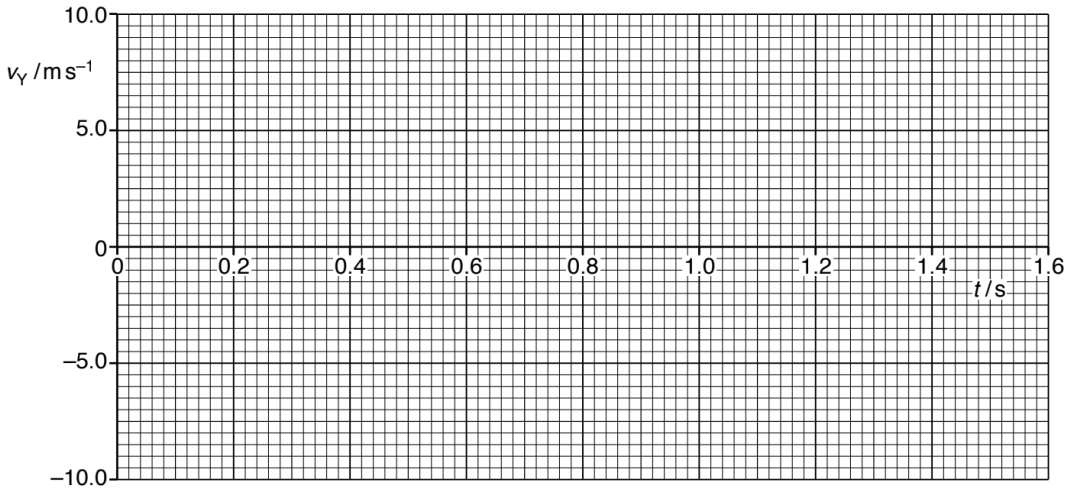


Fig. 1.2

[2]

Mark Scheme (7 marks):

List the known quantities:

- Horizontal displacement, x = 24 m
- Time, t = 1.5 s
- Angle of projection, θ = 28°

(i) Calculate the initial horizontal component v_x of the velocity of the ball:

- $v = \frac{x}{t}$
- $v_x = \frac{24}{1.5}$ = 16 m s^{-1} *[1 mark]*

(ii) Show that the initial vertical component v_Y of the velocity of the ball is 8.5 m s^{-1}:

Method 1: Using the tan function

- $\tan 28° = \frac{v_Y}{v_x}$ *[1 mark]*
- $v_Y = 16 \tan 28°$
- $v_Y = 8.507 \text{ m s}^{-1} = 8.5 \text{ m s}^{-1}$ 2 s.f. *[1 mark]*

OR

Method 2: Resolve v in both directions

- $v_x = v \cos 28°$ ***AND*** $v_Y = v \sin 28°$ *[1 mark]*
- $v_Y = v_x \times \frac{\sin 28°}{\cos 28°} = 16 \times \frac{\sin 28°}{\cos 28°}$

- $v_Y = 8.507 \text{ m s}^{-1} = 8.5 \text{ m s}^{-1}$ 2 s.f. *[1 mark]*

(iii) Calculate the time taken for the ball to reach its maximum height above the ground:

List the known quantities:

- Initial vertical velocity, $u \uparrow = 8.5 \text{ m s}^{-1}$
- Final velocity at maximum height, $v = 0 \text{ m s}^{-1}$
- Vertical acceleration, $a = g = 9.81 \text{ m s}^{-2}$
- Time = t

Select the correct equation of motion and substitute the known quantities:

- $v = u + at$ *[1 mark]*
- $0 = 8.5 + -9.81 \times t$
- $t = \frac{8.5}{9.81} = 0.866 \text{ s}$
- $t = 0.87 \text{ s}$ *[1 mark]*

(iv) On Fig. 1.2, sketch the variation with time t of v_y for the time until the ball hits the wall:

- The known co-ordinates are:
  - t , v_Y
  - (0, 8.5)
  - (0.87, 0)
  - The ball hits the wall at $t = 1.5$ s



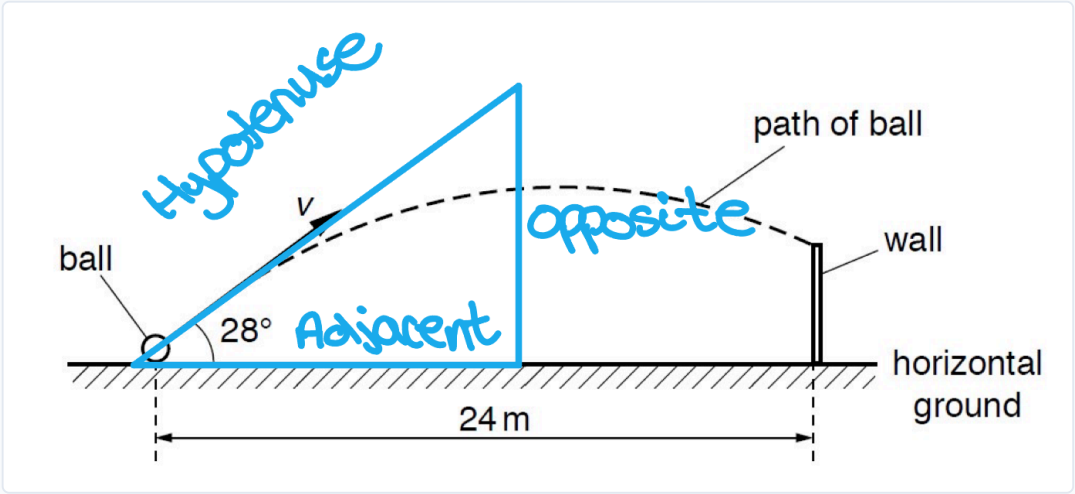
Fig. 2.2

- Straight line drawn from positive v_Y at $t = 0$ to negative v_Y at $t = 1.5$ s *[1 mark]*

- Line starts at $0, \quad 8.5$, passes through $0.87, \quad 0$ and finishes at t = 1.5 s *[1 mark]*

[Total: 7 marks]

- *Draw a **triangle** and label as below, to resolve for v as the **quickest** way to do part (ii)*



Part (c) [4 marks]

(i) Calculate the maximum height above the ground of the ball in **(b)**.

maximum height = ... m [2]

(ii) The maximum gravitational potential energy of the ball above the ground is 22 J.

Calculate the mass of the ball.

mass = ... kg [2]

Mark Scheme (4 marks):

List the known quantities:

- Initial vertical velocity, $u \uparrow$ = 8.5 m s^{-1}
- Final velocity, v = 0 m s^{-1}
- Acceleration of free fall, a = −9.81 m s^{-2}
- Time to maximum height, t = 0.87 s (from part bii)
- Maximum height = s
- Maximum gravitational potential energy of ball, E = 22 J

(i) Calculate the maximum height above the ground of the ball in **(b)**:

- SUVAT: $v^2 = u^2 + 2as$ *[1 mark]*
- $0^2 = 8.5^2 + 2 \times -9.81 \times s$

- $s = \frac{-8.5^2}{2 \times -9.81}$
- $s = 3.68 \text{ m} = 3.7 \text{ m}$ *[1 mark]*

(ii) Calculate the mass of the ball m:

- $\Delta E = mg\Delta h \quad \Rightarrow \quad m = \frac{\Delta E}{g\Delta h}$
- $m = \frac{22}{9.81 \times 3.7}$ *[1 mark]*
- $m = 0.606 \text{ kg} = 0.61 \text{ kg}$ *2 s.f. [1 mark]*

[Total: 4 marks]

- *For part (i), since you have all 5 quantities, you could use any of the following to gain both marks as they lead to the same final answer:*
  - $s = ut + \frac{1}{2}at^2 = 8.5 \times 0.87 + \frac{1}{2}\text{-}9.81 \times 0.87^2$
  - $s = vt - \frac{1}{2}at^2 = 0 - \frac{1}{2}\text{-}9.81 \times 0.87^2$
  - $s = \frac{1}{2}u + vt = \frac{1}{2} \times 8.5 \times 0.87$

Part (d) [1 mark]

A ball of greater mass is kicked with the same velocity as the ball in **(b)**.

Air resistance is still negligible.

State and explain the effect, if any, of the increased mass on the time taken by the ball to reach its maximum height.

Mark Scheme (1 marks):

If the mass is increased, the time taken to reach a maximum height will:

- Stay the same / not be affected

This is because:

- The acceleration (of free fall) is the same / constant for all objects on Earth

  OR

- The acceleration (of free fall) is not dependent on mass, so it has no effect on the time taken; *[1 mark]*

[Total: 1 mark]

Question 6 Medium [18 marks]

Part (a) [6 marks]

A boy standing on a cliff shoots a pellet from his pellet gun vertically upwards. The variation of displacement *s* of the pellet with time *t* is shown in Fig. 1.1.

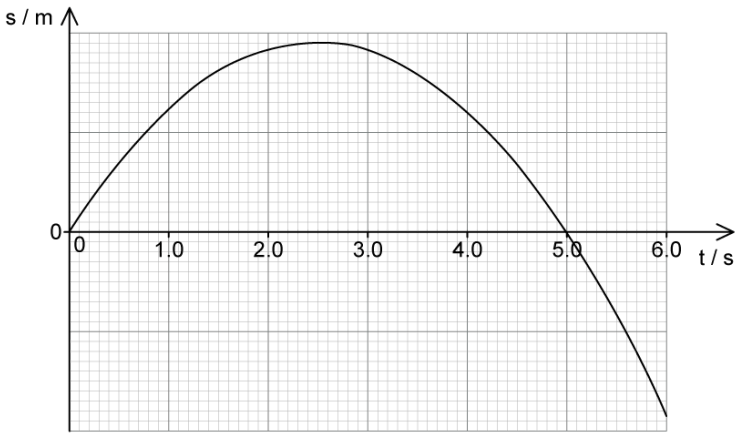


Fig. 1.1

(i) Determine the magnitude of the initial velocity of the pellet, at t = 0 s.

[3]

(ii) Determine the magnitude of the velocity of the pellet at t = 6.0 s.

[2]

(iii) State an assumption made in these calculations.

[1]

Mark Scheme (6 marks):

(i) Determine the magnitude of the initial velocity of the pellet:

Consider the motion from launch $t = 0$ to the maximum height $t = 2.5$ s:

- Initial velocity = u
- Final velocity, v = 0
- Acceleration, a = −9.81 m s^{-2}
- (From Fig. 1.1) time taken, t = 2.5 s

Select an appropriate SUVAT equation to calculate u:

- SUVAT: $v = u + at$ *[1 mark]*
- Substitute: 0 = u − (9.81 × 2.5) *[1 mark]*

- Rearrange & calculate: u = 24.5 m s^{-1} *[1 mark]*

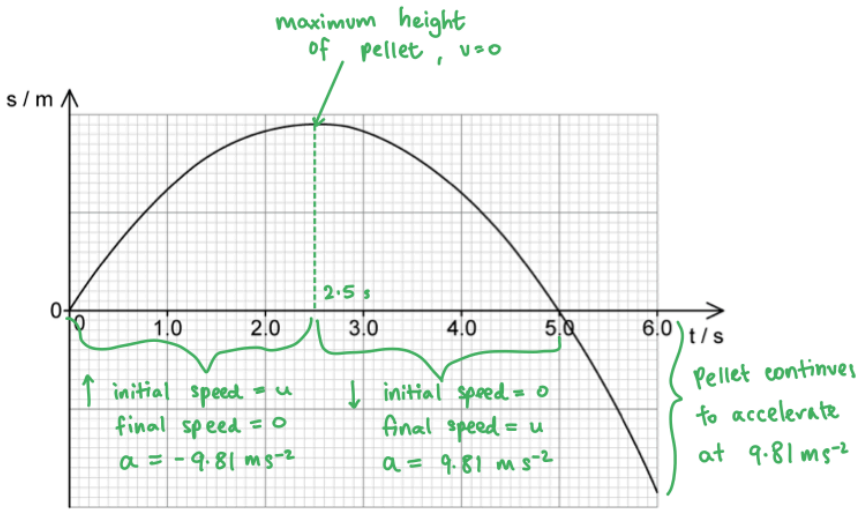
(ii) Determine the final velocity of the pellet:

Consider the motion from the maximum height $t = 2.5$ s to the final position $t = 6.0$ s:

- Initial velocity, u = 0
- Final velocity= v
- Acceleration, a = 9.81 m s^{-2}
- Time taken, t = 6.0 − 2.5 = 3.5 s

Select an appropriate SUVAT equation to calculate v:

- SUVAT: $v = u + at$
- Substitute: v = 0 + (9.81 × 3.5) *[1 mark]*
- Calculate: v = 34.3 m s^{-1} *[1 mark]*



(iii) The assumption made in both calculations is:

- There is no / minimal air resistance **OR** the pellet has constant acceleration; *[1 mark]*

[Total: 6 marks]

Part (b) [3 marks]

On Fig. 1.2, sketch the variation of velocity v of the pellet with time t.

Include your values from **(a)** on the axes.

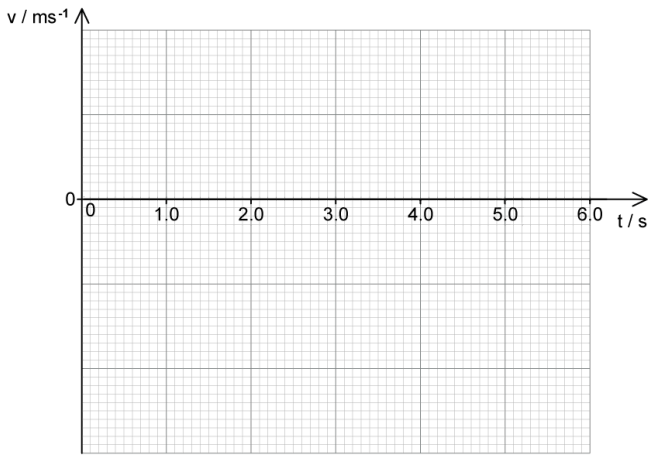
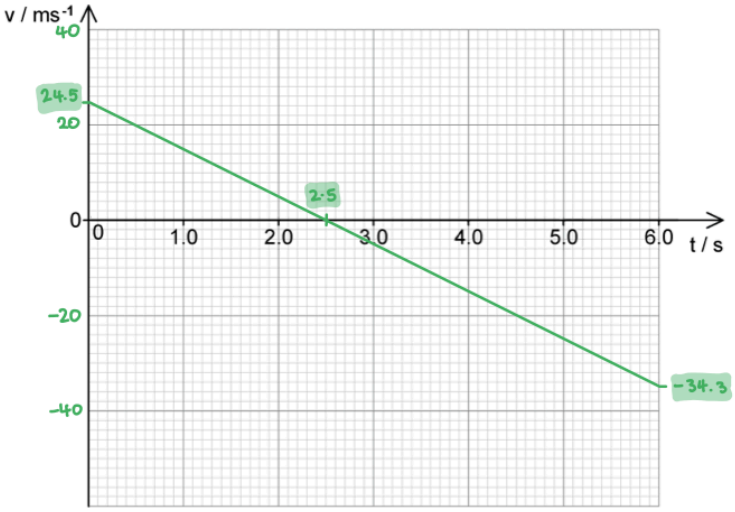


Fig. 1.2

Mark Scheme (3 marks):

The variation of velocity v of the pellet with time t is as follows:



- Straight line starting at a positive value of v **AND** negative gradient; *[1 mark]*
- Line passes time axis at t = 2.5 s (when v = 0); *[1 mark]*
- Values from (a) correctly labelled i.e. 24.5 m s^{-1} at t = 0 and −34.3 m s^{-1} at t = 6.0 s; *[1 mark]*

[Total: 3 marks]

Part (c) [4 marks]

Using Fig. 1.2

(i) Determine the maximum height reached by the pellet.

[2]

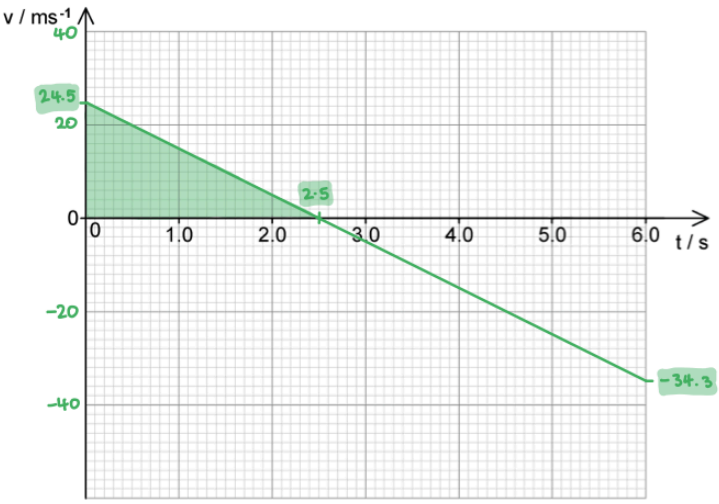
(ii) Show that the final position of the pellet after 6.0 s is about 30 m below its initial position.

[2]

Mark Scheme (4 marks):

(i) Determine the maximum height reached by the pellet:

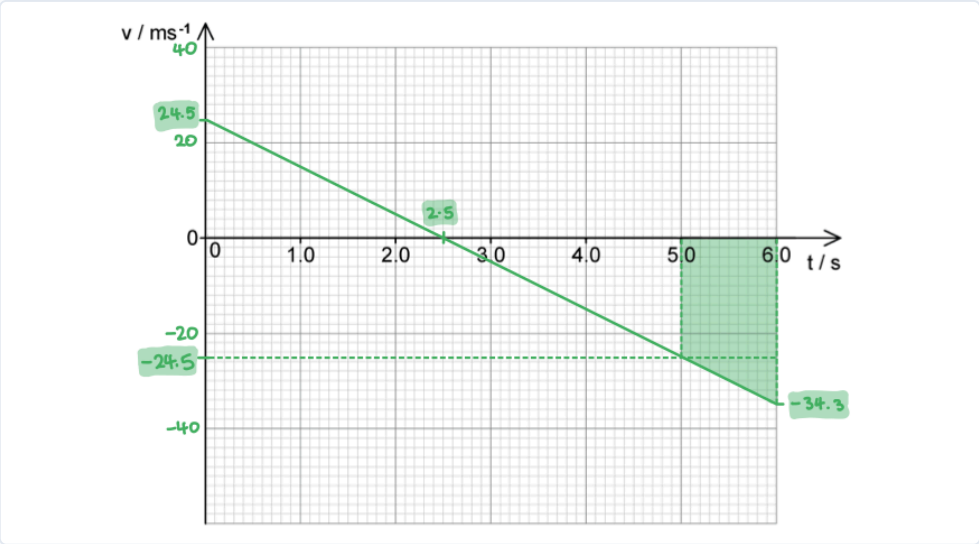
- The maximum height is reached when $v = 0$ at $t = 2.5$ s
- Displacement = area under the velocity-time graph



- Area = $\frac{1}{2} \times 24.5 \times 2.5$ *[1 mark]*
- Maximum height = 30.625 = 30.6 m *[1 mark]*

(ii) Show that the final position of the pellet after 6.0 s is about 30 m below its initial position:

- After the pellet reaches the maximum height of 30.6 m, it falls 30.6 m back to the starting point and then travels in the negative direction as it falls towards the ground under the cliff edge
- The pellet passes the starting point at $t = 5.0$ s
- So, the final displacement is the displacement between 5.0 and 6.0 s



- Area of triangle = $\frac{1}{2} \times 6.0 - 5.0 \times 24.5 - 34.3$
- Area of rectangle = $6.0 - 5.0 \times -24.5$
- Total area = $-\frac{1}{2}\ 34.3 - 24.5 + 24.5$ *[1 mark]*
- Final position = −29.4 m *[1 mark]*

[Total: 4 marks]

For part (ii), you could also calculate the total displacement between 2.5 and 6.0 s and then subtract your answer from (i), this would give the same result

*The minus sign is very important in part (ii) because the question asks you to show that the final position is **below the starting point**. This means that the pellet is travelling in the **opposite direction** to its initial direction of motion, which is directly down from its starting point, hence the need for the minus sign*

Part (d) **[5 marks]**

An identical pellet gun is fired horizontally off the edge of the cliff at the exact same time the other gun is fired vertically upwards. The vertically fired pellet is in motion for exactly 6.0 s before hitting the ground below the edge of the cliff.

Both pellets have the same initial speed.

(i) State which pellet hits the ground first.

[1]

(ii) Calculate the time difference between the pellets hitting the ground.

[4]

Mark Scheme (5 marks):

(i) The pellet which hits the ground first is:

- The horizontally fired pellet; *[1 mark]*

(ii) Calculate the time difference between the pellets hitting the ground:

List the known quantities:

- Height of cliff, s = 29.4 m
- Initial vertical velocity, u = 0
- Acceleration, a = 9.81 m s^{-2}
- Time taken = t

Select an appropriate SUVAT equation to calculate u:

- SUVAT: $s = ut + \frac{1}{2}at^2$ *[1 mark]*
- Substitute & rearrange: $29.4 = 0 + \frac{1}{2} \times 9.81 \times t^2 \quad \Rightarrow \quad t = \sqrt{\frac{2 \times 29.4}{9.81}}$ *[1 mark]*
- Time of flight of horizontal pellet: t = 2.448 = 2.4 s *[1 mark]*
- Time between pellets hitting the ground = 6.0 − 2.4 = 3.6 s *[1 mark]*

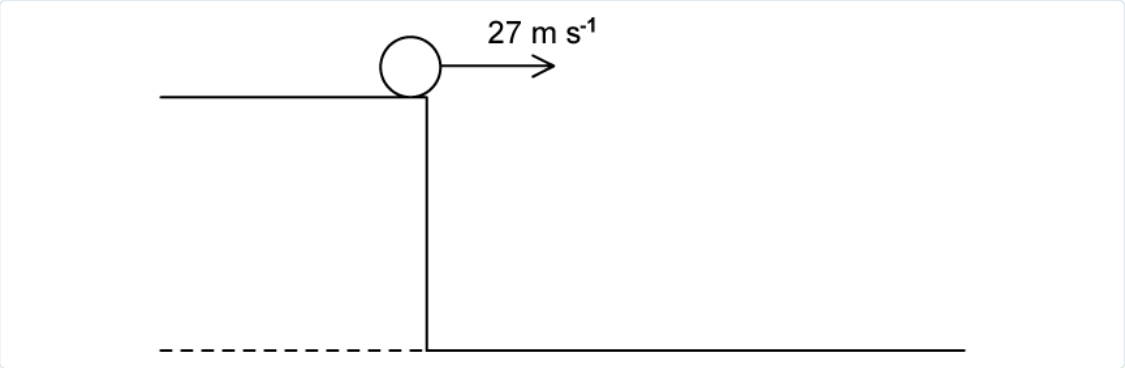
[Total: 5 marks]

Question 7 Medium [11 marks]

Part (a) [4 marks]

A ball is projected horizontally at 27 m s^{-1} from a vertical cliff. It travels a horizontal distance of 40 m before hitting the ground.

Assume that air resistance is negligible.



Calculate the vertical velocity of the ball just before it hits the ground.

Mark Scheme (4 marks):

Calculate the vertical velocity of the ball just before it hits the ground:

List the known quantities:

- Horizontal displacement, s_x = 40 m
- Horizontal initial velocity, u_x = 27.0 m s^{-1}
- Horizontal acceleration, a_x = 0
- Vertical initial velocity, v_y = 0
- Vertical acceleration, $a_y = g$ = 9.81 m s^{-2}

Calculate the time taken to hit the ground, *t*:

- SUVAT: $s = ut + \frac{1}{2}at^2 \quad \Rightarrow \quad s_x = u_x t$
- $t = \frac{s_x}{u_x} = \frac{40}{27}$ *[1 mark]*
- *t* = 1.481 s *[1 mark]*

Calculate the vertical final velocity:

- SUVAT: $v = u + at \quad \Rightarrow \quad v_y = gt$
- *v* = 9.81 × 1.481 *[1 mark]*
- *v* = 14.53 = 15 m s^{-1} *[1 mark]*

[Total: 4 marks]

*Remember you can only use speed = $\frac{distance}{time}$ for objects moving at **constant** velocity (0 acceleration) or if acceleration a is constant as long as **average speed** is used.*

Part (b) [2 marks]

Calculate the height of the cliff.

Mark Scheme (2 marks):

Calculate the height of the cliff:

List the known quantities:

- Vertical acceleration, $a_y = g$ = 9.81 m s^{-2}
- Vertical initial velocity, u_y = 0
- Time taken, t = 1.481 s (from **(a)**)

Substitute values into SUVAT equation to calculate s_y:

- $s = ut + \frac{1}{2}at^2 \quad \Rightarrow \quad s = \frac{1}{2}gt^2$
- $s_y = \frac{1}{2} \times 9.81 \times (1.481)^2$ *[1 mark]*
- s_y = 10.758 = 11 m (2 s.f.) *[1 mark]*

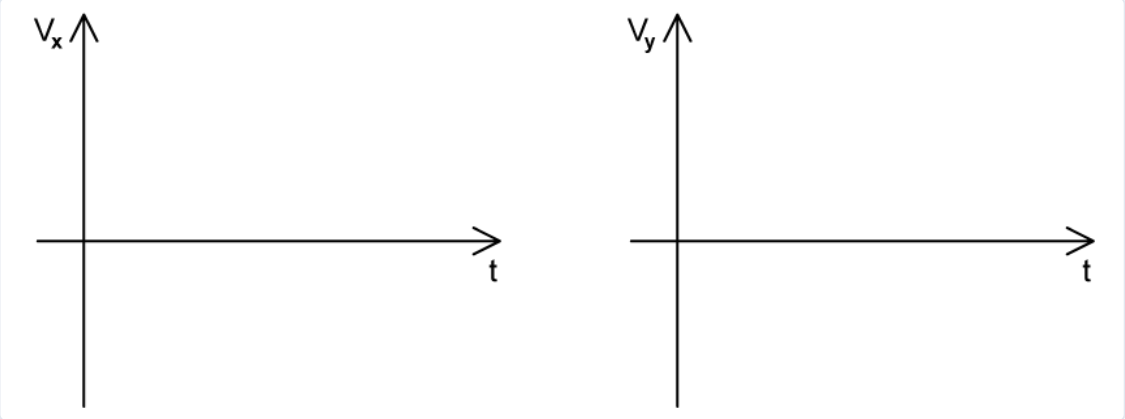
[Total: 2 marks]

This mark scheme has assumed that the downwards direction is positive, it could also be negative, meaning g = –9.81 m s^{-2} which would make the vertical displacement –11 m. This is fine, as long as the positive and negative directions are consistent throughout the question.

Part (c) [3 marks]

Sketch the graphs to show how the horizontal and vertical components of the velocity of the ball, v_x and v_y change with time t just before the ball hits the ground.

Label any appropriate values on the axes.

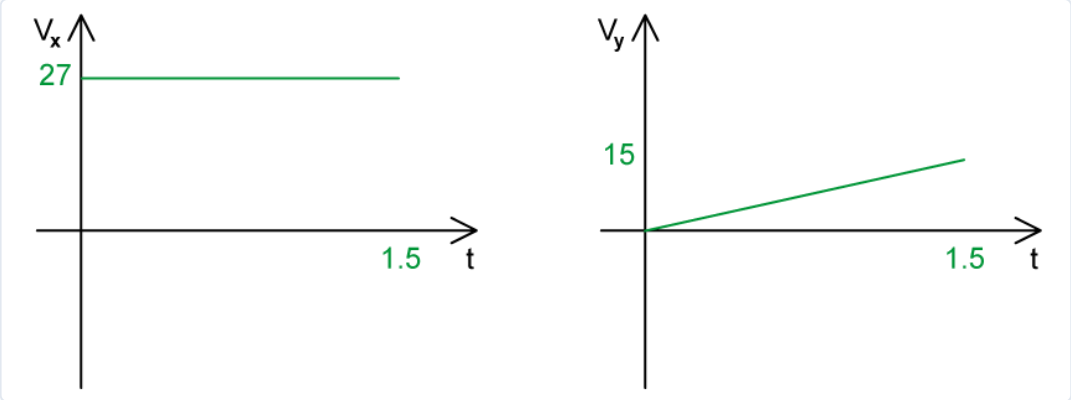


Mark Scheme (3 marks):

Sketch the graphs to show the variation of vertical and horizontal velocity with time:

List the known quantities:

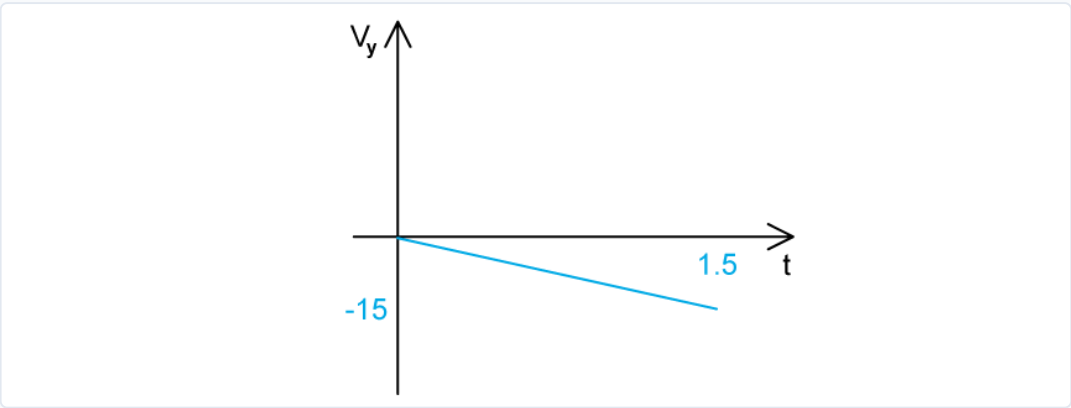
- Time taken, t = 1.481 s = 1.5 s (from **(a)**)
- Horizontal velocity, v_x = 27 m s^{-1}
- Maximum vertical velocity, v_y = 15 m s^{-1} (from **(a)**)



- Horizontal line for v_x – t graph; *[1 mark]*
- Straight line upwards for v_y – t graph; *[1 mark]*
- Values for v_x, v_y and t all labelled correctly; *[1 mark]*

[Total: 3 marks]

The diagonal line for the v_y – t graph can be downwards as well as upwards, this would be the case if you have considered the downwards direction as negative, instead of positive. The graph would look like this:



This is fine, as long as you are consistent with the positive and negative signs throughout the question.

Part (d) [2 marks]

Calculate the resultant velocity of the ball just before it hits the ground.

Mark Scheme (2 marks):

Calculate the resultant velocity of the ball:

List the known quantities:

- Horizontal velocity, v_x = 27 m s^{-1}
- Maximum vertical velocity, v_y = 15 m s^{-1}

Calculate the resultant velocity, *v*:

- $v = \sqrt{{v_x}^2 + {v_y}^2} = \sqrt{15^2 + 27^2}$ *[1 mark]*
- *v* = 31 m s^{-1} *[1 mark]*

[Total: 2 marks]

Up until this point, velocity has always been considered in terms of its horizontal and vertical components. At this point, the resultant velocity is always calculated using Pythagoras (which is the case for all resultant vectors split into horizontal and vertical components)!

Question 8 Medium [19 marks]

Part (a) [8 marks]

A particle moves in a straight line with a constant acceleration, *a*. The initial velocity of the particle at *t* = 0 is *u*. After a further time *t*, its velocity has increased to *v*.

(i) Using the definition of acceleration, show that

$$v = u + at$$

[2]

(ii) Using the definition of average velocity, show that

$$s = ut + \frac{1}{2}at^2$$

[3]

(iii) Using these two equations, show that

$$v^2 = u^2 + 2as$$

[3]

Mark Scheme (8 marks):

(i) Derive $v = u + at$

- Acceleration is defined as
  - $a = \frac{\text{change in velocity}}{\text{time taken}} = \frac{\Delta v}{\Delta t}$ *[1 mark]*
- The change in velocity in this case is from *u* to *v*
- Substitute in $\Delta v = v - u$:
  - $a = \frac{v - u}{t}$ *[1 mark]*
- Rearrange into the required form:
  - $at = v - u$
  - $v = u + at$

(ii) Derive $s = ut + \frac{1}{2}at^2$

- Average velocity is defined as:
  - $\bar{v} = \frac{\text{distance travelled}}{\text{time taken}} = \frac{s}{t}$ *[1 mark]*
- The average velocity in this case is the average of *u* and *v*
  - $\bar{v} = \frac{u + v}{2}$
- Equate the expressions:
  - $\frac{s}{t} = \frac{u + v}{2}$

- $s = \frac{u + v}{2}t$ *[1 mark]*

- Substitute in $v = u + at$:
  - $s = \frac{u + u + at}{2}t$ *[1 mark]*

- Simplify:
  - $s = \frac{2ut + at^2}{2}$
  - $s = ut + \frac{1}{2}at^2$

(iii) Derive $v^2 = u^2 + 2as$

- Rearrange:
  - $v = u + at$
  - $t = \frac{v - u}{a}$

- Substitute:
  - $s = ut + \frac{1}{2}at^2$
  - $s = u\frac{v - u}{a} + \frac{1}{2}a\frac{v - u}{a}^2$ *[1 mark]*

- Multiply both sides by 2*a*:
  - $s = \frac{uv - u}{a} + \frac{av - u^2}{2a^2}$
  - $2as = 2uv - u + v - u^2$ *[1 mark]*

- Expand and simplify:
  - $2as = 2uv - 2u^2 + v^2 - 2uv + u^2$
  - $2as = v^2 - u^2$ *[1 mark]*

- Rearrange into the required form:
  - $v^2 = u^2 + 2as$

[Total: 8 marks]

Part (b) **[4 marks]**

A student investigates the motion of a light ball bearing as it falls from rest alongside a vertical 100 cm scale.

The student sets up a camera to take flash photographs of the ball at 0.05 s intervals, as shown in Fig. 1.1.

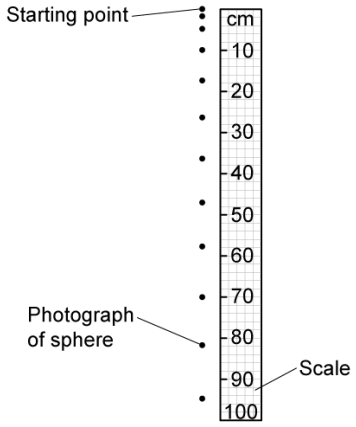


Fig. 1.1

The first photograph is taken at time $t = 0$.

Using Fig. 1.1, describe and explain the motion of the ball

(i) in the interval $t = 0$ to $t = 0.35$ s

[2]

(ii) in the interval $t = 0.35$ to $t = 0.55$ s.

[2]

Mark Scheme (4 marks):

(i) In the interval $t = 0$ to $t = 0.35$ s:

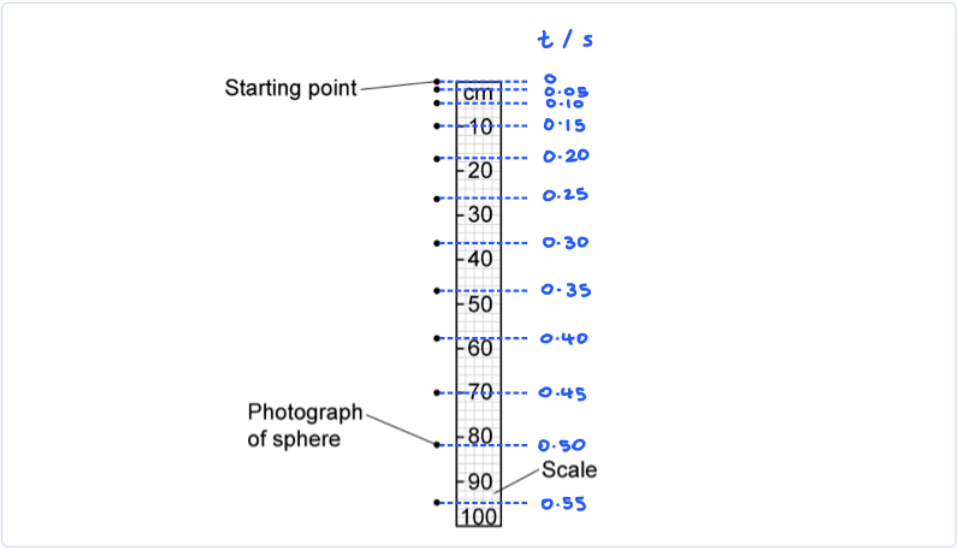
- The ball accelerates; *[1 mark]*
- (Because) the separation of dots becomes larger / increases between each interval; *[1 mark]*

(ii) In the interval $t = 0.35$ to $t = 0.55$ s:

- The ball reaches a constant speed; *[1 mark]*
- (Because) the separation of dots becomes constant / does not continue to increase; *[1 mark]*

[Total: 4 marks]

It is very useful to annotate the diagram with the times to see that there is a clear distinction between the two time intervals in terms of the gaps between each dot



Part (c) [4 marks]

The 100 cm scale is suspended 160 cm above the ground.

Determine the total time of flight for the ball after it is dropped from the starting point of the 100 cm scale.

Mark Scheme (4 marks):

Determine the total time of flight for the ball:

- The ball falls from the starting point to 95 cm in 0.55 s
- Therefore, after 0.55 s, it falls 160 + 5 = 165 cm *[1 mark]*
    - Hence, distance, s = 165 cm
- At constant speed, distance travelled in 0.1 s = 25 cm **OR** in 0.05 s = 12.5 cm *[1 mark]*
    - Hence, speed, v = 250 cm s^{-1}

  Acceptable range for distance travelled in 0.1 s = 24 to 26 cm or distance travelled in 0.05 s = 11 to 13 cm

Calculate the time during the fall at constant speed: speed 220 - 260

- Speed equation: $v = \frac{s}{t} \Rightarrow t = \frac{s}{v}$
- $t = \frac{165}{250} = 0.66$ s *[1 mark]*

  Acceptable range for t = 0.63 to 0.75 s
- Total time of flight = 0.55 + 0.66 = 1.21 s *[1 mark]*

[Total: 4 marks]

Part (d) [3 marks]

The student repeats the experiment with a lead sphere that falls with constant acceleration and does not reach a constant speed.

Determine the number of photographs, including the photograph at t = 0, that will be observed against the 100 cm scale.

Mark Scheme (3 marks):

Determine the number of photographs that will be observed against the 100 cm scale:

List the known quantities:

- Initial speed, u = 0
- Distance, s = 100 cm = 1 m
- Acceleration, $a = g$ = 9.81 m s^{-2}
- time = t

Use a SUVAT equation to find time t:

- $s = ut + \frac{1}{2}at^2 \quad \Rightarrow \quad s = \frac{1}{2}gt^2$
- $t = \sqrt{\frac{2 \times 1}{9.81}}$ *[1 mark]*
- t = 0.45 s *[1 mark]*

Determine the number of photographs:

- The camera takes a photo every 0.05 s
- Number of photographs = $\frac{0.45}{0.05} + 1$
- Number of photographs = 10 *[1 mark]*

[Total: 3 marks]

Question 9 Medium [9 marks]

Part (a) [3 marks]

An archer fires an arrow towards a target as shown in Fig. 1.1.

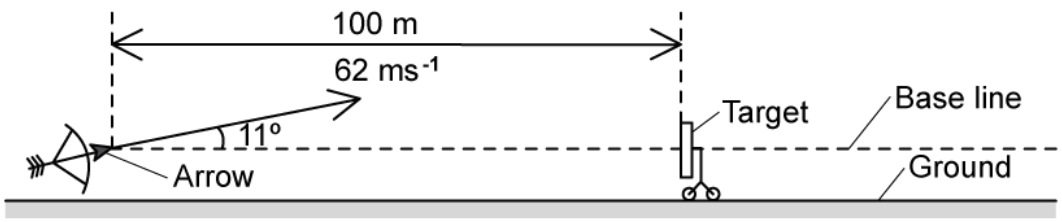


Fig. 1.1 (not to scale)

The centre of the target is at the same height as the initial position of the arrow.

The target is a distance of 100 m from the arrow. The arrow has an initial velocity of 62 m s^{-1} and is fired at an angle of 11° to the horizontal.

Describe how the vertical and horizontal components of the arrow's velocity change from when it is fired until it reaches its maximum height.

Assume that the effect of air resistance is negligible.

Mark Scheme (3 marks):

The vertical component of velocity:

- Decreases with height; *[1 mark]*
- At maximum height, the (vertical) velocity is zero; *[1 mark]*

The horizontal component of velocity:

- Does not change; *[1 mark]*

[Total: 3 marks]

Part (b) [3 marks]

Show that the time taken for the arrow to reach its maximum height is about 1.2 s.

Mark Scheme (3 marks):

Show that the time taken for the arrow to reach its maximum height is about 1.2 s:

List the known quantities:

- Initial velocity, u = 62 m s^{-1}

- Final vertical velocity, $v = 0$
- Acceleration, $a = -g = -9.81$ m s^{-2}
- Time to reach maximum height = t

Resolve the initial velocity into its vertical component:

- $u = 62 \sin(11°)$ *[1 mark]*

Substitute into the appropriate SUVAT equation and evaluate:

- SUVAT: $v = u + at$
- $0 = 62 \sin(11°) - gt$ *[1 mark]*
- $t = \frac{62 \sin 11°}{9.81} = 1.21$ s *[1 mark]*

[Total: 3 marks]

Part (c) **[3 marks]**

Calculate the horizontal distance, measured along the base line, by which the arrow misses the target.

Mark Scheme (3 marks):

Calculate the horizontal distance by which the arrow misses the target:

List the known quantities:

- Initial velocity, $u = 62$ m s^{-1}
- Horizontal acceleration, $a = 0$
- Time to reach maximum height, $t = 1.21$ s (from **(b)**)
- Horizontal distance = s

Determine the total time of flight:

- Time of flight, $t = 2 \times 1.21 = 2.42$ s *[1 mark]*

Resolve the initial velocity into its horizontal component:

- $u(\rightarrow) = 62 \cos(11°)$

Substitute into the chosen SUVAT equation and evaluate:

- SUVAT: $s = ut + \frac{1}{2}at^2 \quad \Rightarrow \quad s = ut$
- $s = 62 \cos(11°) \times 2.42 = 147$ m *[1 mark]*

Determine the distance by which the arrow misses the target

- Horizontal distance = 147 – 100 = 47 m *[1 mark]*

[Total: 3 marks]

- *Always write down all your working, even if you get a bit lost there are lots of opportunities to gain marks even if your final answer is incorrect*
    - *As long as you multiplied your time from part (b) by 2, and correctly followed the rest of the steps, then you would likely score all the marks*
    - *If you didn't multiply your time by 2, you would only be able to score 1 mark if 1.2 × 62 cos (11°) were seen*
- *Notice that since you were already awarded a mark in part (b) for resolving the velocity component, you wouldn't be awarded another mark for doing it correctly in this part*

Question 10 Hard [11 marks]

Part (a) [4 marks]

A motorcyclist practices a stunt on a ramp inclined such that they leave it at an angle of 25° and 3 m high as shown in Fig. 1.1.

The other ramp is 24 m away and has a height of 1 m, as shown.

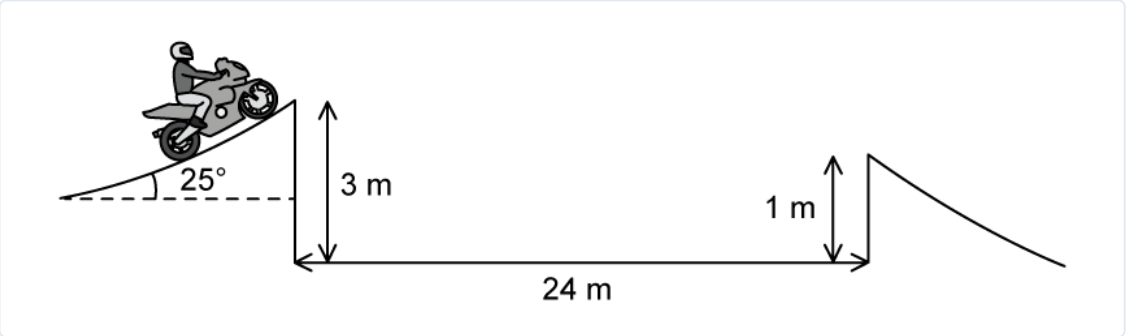


Fig. 1.1

Calculate the minimum initial velocity the motorcyclist requires in order to reach the other ramp.

Mark Scheme (4 marks):

Calculate the minimum initial velocity the motorcyclist requires in order to reach the other ramp:

List the known quantities:

- Angle of incline, θ = 25°
- Height of first ramp = 3 m
- Height of second ramp = 1 m
- Horizontal distance, x = 24 m

Determine an expression for the initial velocity, u in terms of time t:

- In the horizontal direction: x = 24 m, $u_h = u \cos 25°$, a = 0, t = ?
- In the horizontal direction: $x = u_h t$
- $x = u \cos 25° t \Rightarrow u = \frac{x}{t \cos 25°}$
- $u = \frac{24}{t \cos 25°}$ *[1 mark]* **[Eq. 1]**

Determine equation for time t from the vertical direction:

- In the vertical direction: Δs = –2 m, $u_v = u \sin 25°$, a = –9.81 m s^{-2}, t = ?
- In the vertical direction: $s = ut + \frac{1}{2}at^2 \Rightarrow \Delta s = u_v t - \frac{1}{2}gt^2$
- $-2 = u \sin 25° t - \frac{9.81}{2}t^2$ *[1 mark]* **[Eq. 2]**

Calculate *t* using simultaneous equations:

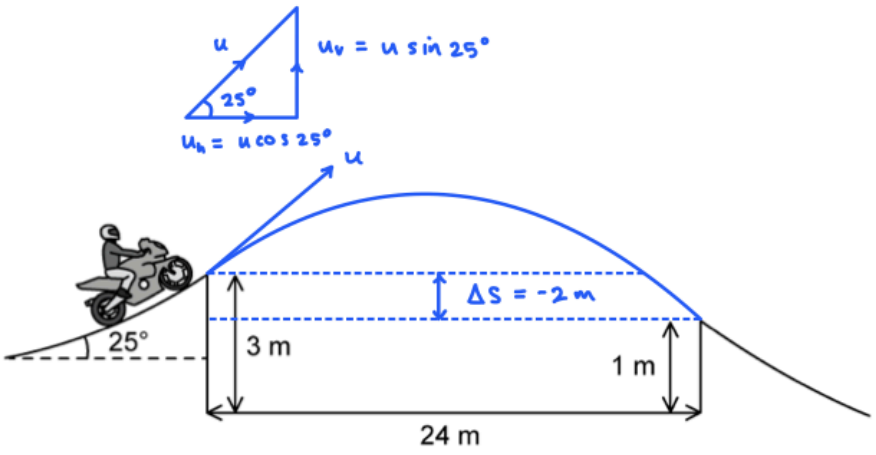
- Substitute u from **[Eq. 1]** into **[Eq. 2]**
- $-2 = \frac{24\ t\ \sin\ 25°}{t\ \cos\ 25°} - \frac{9.81}{2}\ t^2$
- $\frac{9.81}{2}\ t^2 = 24\ \tan\ 25° + 2 \Rightarrow t^2 = \frac{48\ \tan\ 25° + 4}{9.81}$
- $t = \sqrt{\frac{48\ \tan\ 25° + 4}{9.81}} = 1.64$ s *[1 mark]*

Calculate initial velocity using **[Eq. 1]**:

- $u = \frac{24}{1.64 \times \cos\ 25°} = 16$ m s^{-1} *[1 mark]*

[Total: 4 marks]

*The key to this kind of problem is to consider the **total vertical displacement** due to the difference in initial and final heights*



Part (b) [4 marks]

The stunt motorcyclist's ultimate goal is to jump across a river and land on the other side. Fig. 1.3 shows the motorcyclist driving off a ramp at the edge of a river.

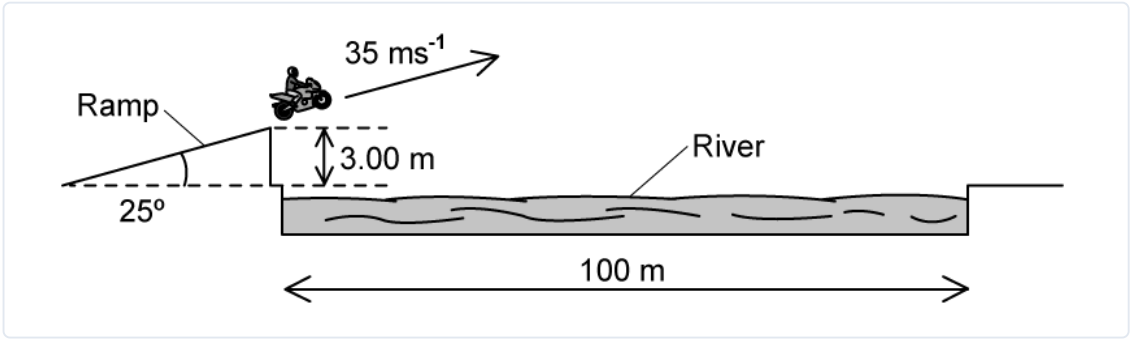


Fig. 1.3

The ramp is at an angle of 25° to the horizontal and the height at the end of the ramp is 3.0 m. The width of the river is 100 m. The initial velocity of the motorcyclist is 35 m s^{-1}.

Deduce whether the motorcyclist lands on the other side of the river.

Mark Scheme (4 marks):

Deduce whether the rider lands on the other side of the river:

List the known quantities:

- Required horizontal distance, x = 100 m
- Vertical height above ground, h = 3.00 m
- Initial velocity, u = 35 m s^{-1}
- Angle of incline of ramp, θ = 25°
- Horizontal acceleration = 0
- Vertical acceleration, a = -g = −9.81 m s^{-2}

Determine the horizontal and vertical components of the initial velocity:

- Resolving horizontally (→): $u_x = 35 \times \cos 25°$ = 31.7 m s^{-1}
- Resolving vertically (↑): $u_y = 35 \times \sin 25°$ = 14.8 m s^{-1}

Determine the time taken to travel 100 m horizontally:

- Horizontal motion: $s = u_x t$
- $t = \frac{100}{31.7}$ = 3.155 s *[1 mark]*

Determine the vertical distance fallen in this time:

- Vertical motion: $s = u_y t + \frac{1}{2}at^2$
- $s = 14.8 \times 3.155 - \frac{1}{2} \times 9.81 \times 3.155^2$ = −2.13 m *[1 mark]*
- Vertical distance fallen = 2.1 m *[1 mark]*

Write a concluding statement:

- When the rider has travelled 100 m horizontally, they have fallen a distance of 2.1 m which is less than 3.0 m, hence the rider lands on the other side of the river; *[1 mark]*

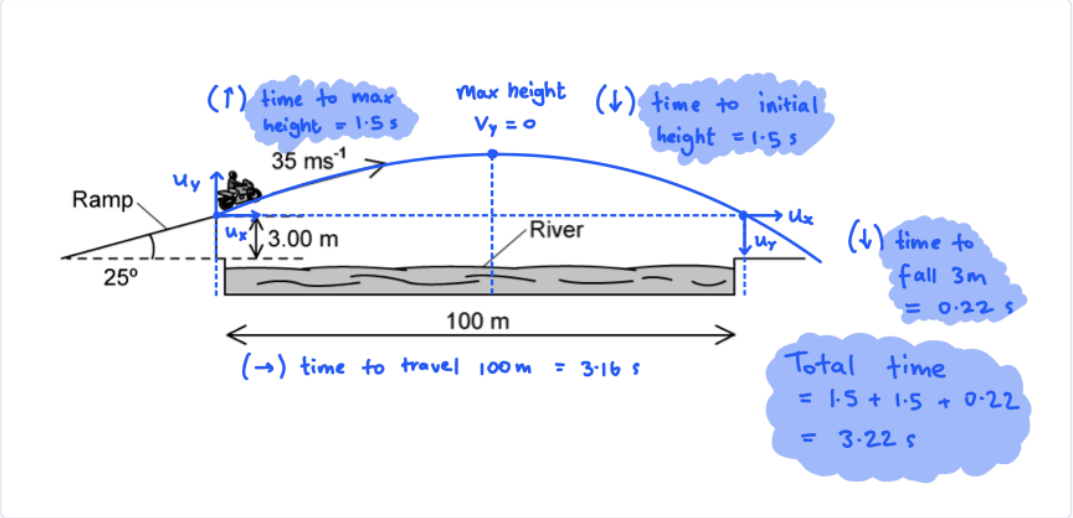
[Total: 4 marks]

The method shown above is by far the simplest way to tackle this question and requires the fewest calculation steps with no need for a quadratic formula

There are other methods which give equally clear results but are more prone to calculation errors along the way

- ***Alternative method 1: compare the times of flight*** *Expected conclusion: it takes the rider 3.16 s to travel 100 m horizontally which is less than 3.22 s, the total time of flight, hence the rider lands on the other side of the river*
- ***Alternative method 2: compare the horizontal distances*** *Expected conclusion: the rider travels a total horizontal distance of 102 m which is greater than 100 m, hence the rider lands on the other side of the river*

It can be useful to draw a clear sketch to help yourself to visualise the situation:



Part (c) [3 marks]

Explain how air resistance would affect the jump.

Mark Scheme (3 marks):

The effect of air resistance on the rider's motion would be:

- Air resistance acts to oppose the motion of the motorcyclist; *[1 mark]*

*Max **one** from:*

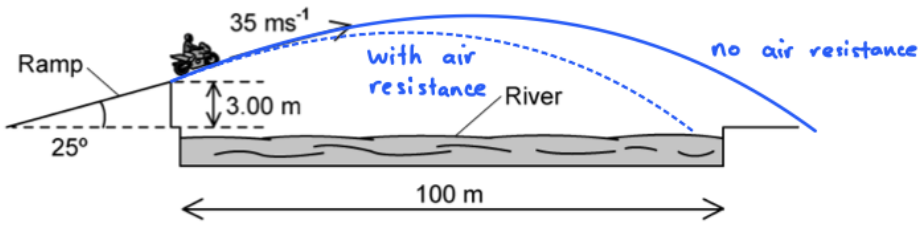
- It decreases the time for which the motorcyclist is in the air; *[1 mark]*
- It causes deceleration in the horizontal direction; *[1 mark]*
- It reduces speed in the horizontal direction; *[1 mark]*
- It reduces the (maximum) height reached by the motorcyclist; *[1 mark]*

*If any mark is gained from the above, max **one** for:*

- It reduces the horizontal distance travelled; *[1 mark]*

[Total: 3 marks]

You could also use a diagram to aid your description - a clearly annotated sketch could earn you some marks


35 ms-1
Ramp
3.00 m
25°
with air resistance
River
no air resistance
100 m

Question 11 Hard [12 marks]

Part (a) [3 marks]

An object is released near the surface of the Moon at time $t = 0$. Fig. 1.1 shows the variation of displacement s with time t of the object from the point of release.

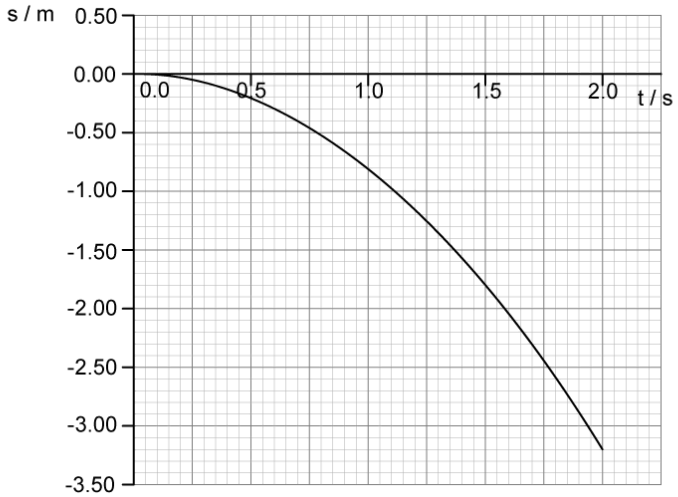


Fig. 1.1

(i) State the significance of the negative values of s.

[1]

(ii) State two assumptions about the motion of the object.

[2]

Mark Scheme (3 marks):

(i) The significance of the negative values of s is that its direction is:

- Upwards / away from the Moon is taken as positive **OR** downwards / towards the Moon is taken as negative; *[1 mark]*

(ii) An assumption about the motion of the object is:

- There is no effect of air resistance; *[1 mark]*
- There is no gravitational effect from the Earth / another body; *[1 mark]*

[Total: 3 marks]

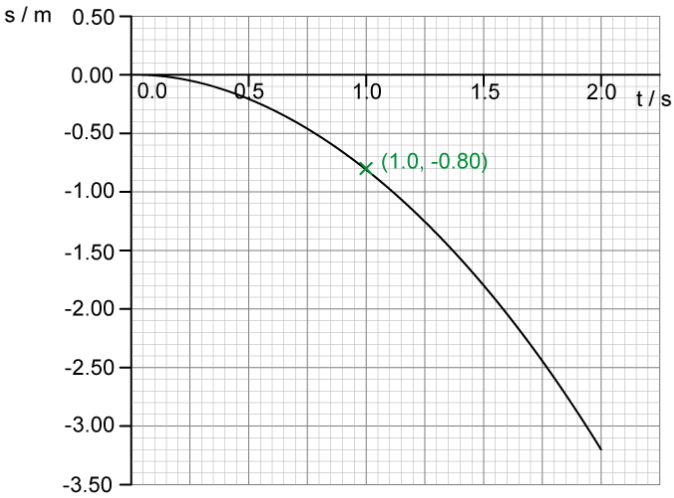
Part (b) [2 marks]

Use the graph in Fig. 1.1 to determine a value for the acceleration of free fall close to the surface of the Moon.

Mark Scheme (2 marks):

Use the graph to determine a value for *g* close to the surface of the Moon:

- Take a point on the curve:



- A suitable point labelled on the graph **OR** coordinates stated, e.g. when t = 1.0 s, s = –0.80 m; *[1 mark]*

Substitute coordinates into the equation of motion to find *a*:

- $s = ut + \frac{1}{2}at^2$
- $-0.80 = 0 + \frac{1}{2} \times a \times 1.0$
- a = (2 × –0.80) = –1.6 m s^{-2} *[1 mark]*

[Total: 2 marks]

Any object accelerating uniformly is described by the equation $s = ut + \frac{1}{2}at^2$

*You should realise that this is **the equation of a graph**, specifically, a displacement-time graph – as shown in this question. This is because* $s \propto -t^2$ *and the graph of* $y \propto -x^2$ *is a 'n' shape.*

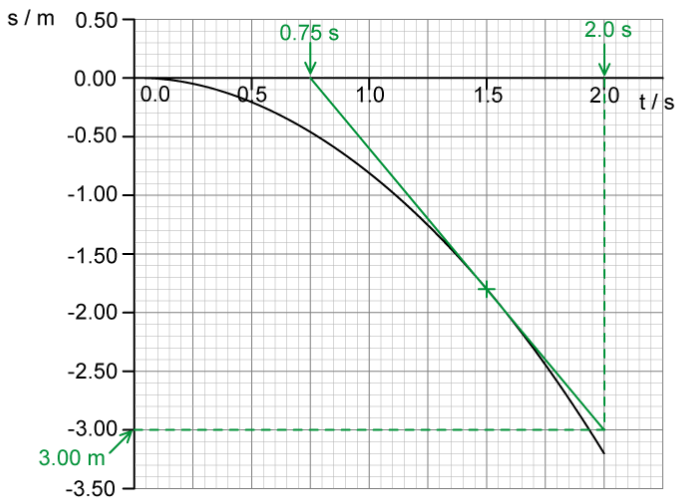
Hence, each coordinate on the graph satisfies this equation for s and t (where a is the acceleration due to free fall and u is the initial velocity)

Part (c) **[3 marks]**

Use Fig. 1.1 to estimate the instantaneous velocity of the object when t = 1.5 s.

Mark Scheme (3 marks):

Use the graph to estimate the instantaneous velocity:



- Tangent drawn at t = 1.5 s *[1 mark]*

Determine the gradient of the tangent:

- Gradient of the tangent = instantaneous velocity at that point
- gradient $= \frac{\Delta s}{\Delta t}$
- gradient $= \frac{-3.00 - 0}{2.0 - 0.75}$ *[1 mark]*
- Instantaneous velocity: $v = -2.4$ m s^{-1} *[1 mark]*

[Total: 3 marks]

*The gradient of a displacement-time graph represents the **velocity** of the object*

*Therefore, the gradient of the tangent at t = 1.5 s estimates the **instantaneous velocity** of the object*

Part (d) **[4 marks]**

(i) On Fig. 1.1, sketch the variation of displacement s with time t if the same object was released close to the surface of the Earth instead.

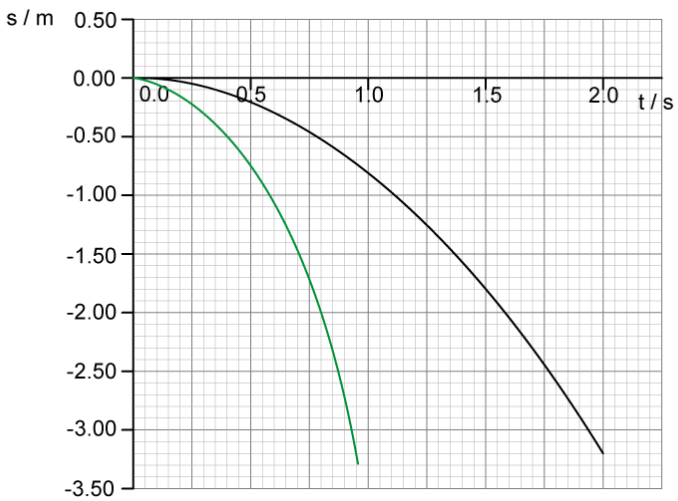
[2]

(ii) Describe and explain the features of your sketch.

[2]

Mark Scheme (4 marks):

(i) Sketch a displacement-time graph that would show the same object released close to the surface of Earth:



- A smooth parabola (judged by eye, as shown); *[1 mark]*
- A line always below the Moon curve (as shown); *[1 mark]*

(ii) Describe and explain the features of the sketch:

- The line is parabolic because the displacement *s* is proportional to t^2 / $s \quad \propto \quad t^2$ *[1 mark]*
- The line is below the Moon curve because the acceleration due to gravity / *g* on Earth is larger than on the Moon; *[1 mark]*

[Total: 4 marks]

If you're ever stuck on a question after part (a), make sure to look at the previous parts for clues to help you answer the question further along. From (b) we calculated the acceleration due to gravity on the moon and this is much less than the Earth. With your knowledge of s-t graphs (the gradient is the velocity) then the rate of change of that gradient (the acceleration) must be bigger for the graph for Earth i.e a steeper curve.

Question 12 Hard [16 marks]

Part (a) [4 marks]

A rock is thrown horizontally off a 175 m high cliff and lands 105 m away.

Calculate the resultant velocity at which it hits the ground, assuming air resistance is negligible.

Mark Scheme (4 marks):

List the known quantities:

- Vertical height, s = 175 m
- Horizontal distance, x = 105 m
- Acceleration due to free-fall, a = 9.81 m s^{-2}

Determine the final vertical velocity, v_y:

- From the data booklet: $v^2 = u^2 + 2as$
- $v_y = \sqrt{0^2 + 2 \times 9.81 \times 175}$
- $v_y = 58.60$ ms^{-1} *[1 mark]*

Calculate the time to fall, t:

- Recall the equation: $v = u + at$
- $t = \frac{v - u}{a}$
- $t = \frac{58.6 - 0}{9.81}$
- $t = 5.97$ s *[1 mark]*

Calculate the horizontal velocity, v_x:

- Recall: $v = \frac{x}{t}$
- $v_x = \frac{105}{5.97}$
- $v_x = 17.59$ m s^{-1} *[1 mark]*

Calculate the resultant velocity, v:

- Use Pythagoras: $a^2 + b^2 = c^2$
- $v = \sqrt{{v_x}^2 + {v_y}^2}$
- $v = \sqrt{17.59^2 + 58.60^2}$
- $v = 61.18$ m s^{-1} *[1 mark]*

[Total: 4 marks]

*Time t is the **same** for the equations representing the horizontal motion or the vertical motion of the ball and this key point is often vital for more complicated projectile motion questions. Anything in projectile motion has a constant horizontal velocity (hence you can use distance = speed × time), but a changing vertical velocity. This is because the acceleration due to gravity is only in the vertical direction.*

Part (b) [3 marks]

The rock hits the ground at an angle θ to the horizontal.

Determine, without calculation, how the angle will change if the stone is thrown in the same way with a greater velocity.

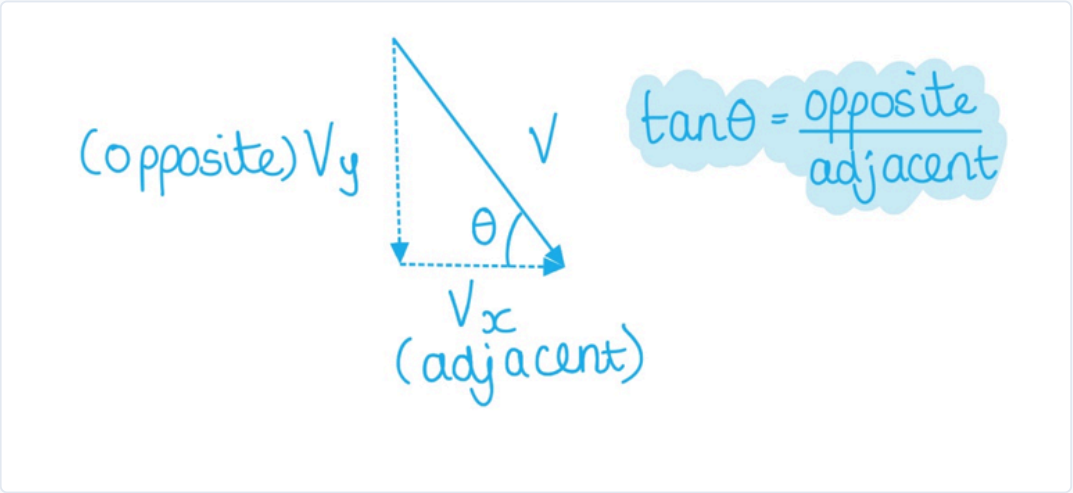
Mark Scheme (3 marks):

If the stone was thrown with a greater horizontal velocity then:

- The adjacent side of the triangle v_x will be bigger; *[1 mark]*
- $\tan \theta = \frac{\text{opp}}{\text{adjacent}} = \frac{v_y}{v_x}$ *[1 mark]*
- So, $\frac{v_y}{v_x}$ will be smaller, hence, θ will be smaller; *[1 mark]*

[Total: 3 marks]

It can be helpful to draw a diagram to show this situation and make it easier to figure out the relationship between the angle and horizontal velocity.



Part (c) [3 marks]

The rock is now attached to a catapult and launched at a castle wall. The rock is launched at an angle of 30° from the horizontal at a speed of 29 m s^{-1}. The castle wall is 50 m away and is 15 m high.

Determine whether the rock makes it over the wall.

Mark Scheme (3 marks):

List the known quantities:

- Angle, θ = 30°
- Speed, u = 29 m s^{-1}
- Horizontal distance, x = 50 m
- Vertical height of the wall, h = 15 m

Calculate time taken to reach the wall, t:

- In the horizontal direction: *Distance = speed × time*
- $x = u_x \times t$ therefore $t = \frac{x}{u_x}$
- $t = \frac{50}{29\cos 30} = 1.99$ s *[1 mark]*

Calculate vertical height of the projectile, s:

- In the vertical direction: s = ?, u = 29sin(30), v = not given, a = –9.81 m s^{-2}, t = 1.99 s
- From the data booklet: $s = ut + \frac{1}{2}at^2$
- $s = (29\sin(30) \times 1.99) + (\frac{1}{2} \times -9.81 \times (1.99)^2) = 9.43$ m *[1 mark]*

Conclusion:

- Since s = 9.43 < h = 15 m, the rock does not make it over the wall; *[1 mark]*

[Total: 3 marks]

If you define the upwards (initial velocity) as positive, then downwards (g) must be negative and vice versa. It doesn't matter which direction you define as positive or negative as long as you are consistent throughout the question

Part (d) **[6 marks]**

The double angle formula is given by sin(2A) = 2sin(A)cos(A).

Justify that, in the absence of air resistance, the maximum range of any projectile that starts and ends at ground level is achieved when it is launched at 45° to the horizontal.

Mark Scheme (6 marks):

Determine the equation for the distance in the horizontal direction:

- $u_x = u\cos(\theta)$
- In the horizonal direction: *distance = speed × time*
- Horizontal distance, $x = u\cos(\theta)\, t$

Determine the equation for time, t:

- In the vertical direction: s = not given, $u_y = u \sin(\theta)$, $v_y = -u \sin(\theta)$, $a = -g$, $t = t$

- Recall: $v = u + at$
- $t = \frac{v - u}{a} = \frac{-u\sin\theta - u\sin\theta}{-g}$ *[1 mark]*
- $t = \frac{2u\sin\theta}{g}$ *[1 mark]*

Substitute the equation for t into the horizontal distance equation:

- $x = u\cos(\theta)\, t = u\cos(\theta)\frac{2u\sin\theta}{g} = \frac{2u^2\cos\theta\sin\theta}{g}$ *[1 mark]*
- Using the double angle formula: sin(2A) = 2sin(A)cos(A)
- $x = \frac{u^2\sin 2\theta}{g}$ *[1 mark]*

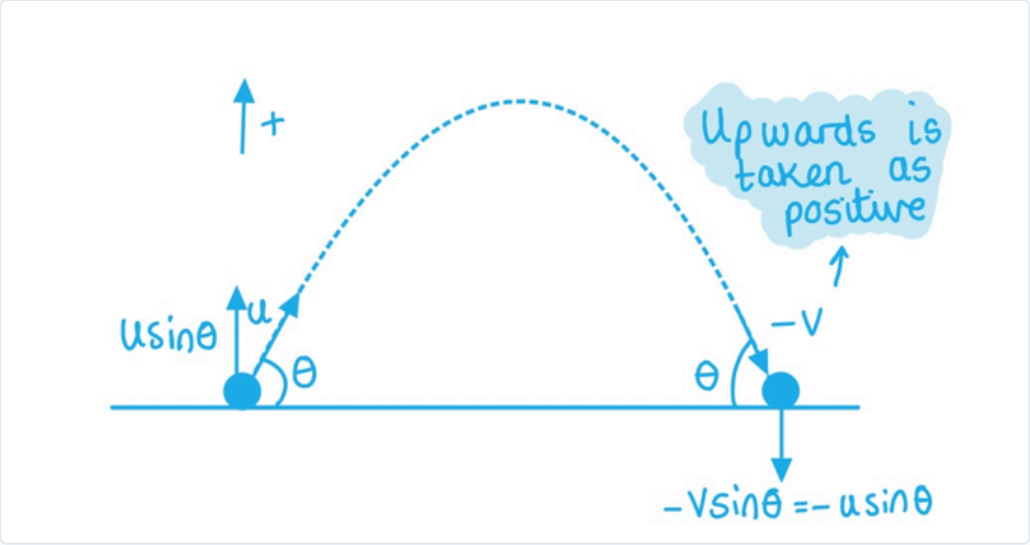
Determine the maximum value of θ:

- $\sin(2\theta) = 1$ therefore, $2\theta = \sin^{-1} 1$ *[1 mark]*
- $2\theta = 90°$
- $\theta = \frac{90}{2} = 45°$ *[1 mark]*

[Total: 6 marks]

Don't be too alarmed if you didn't manage this on your first go, it would be an incredibly harsh exam question. The key to this derivation is the final and initial vertical velocities are equal in magnitude and opposite in direction for a projectile starting and ending at the ground level.

The final vertical velocity, v, will equal the initial vertical velocity u but in the ***opposite direction*** *because of the symmetry of the parabola. Remember, that the greatest value of sin(θ) is equal to 1.*



You will not be expected to remember the double angle formula in your A–level physics exam. Any formulas that are not required will be given in the question, as is done here, which you will be expected to use.

Question 13 Hard [8 marks]

Part (a) [4 marks]

A bus driver drives with a constant acceleration of 5.0 m s^{-2} from an initial velocity of 3.0 m s^{-1}. After 2.5 seconds, they slow down to stop at a set of traffic lights at 4.0 m s^{-2}, eventually coming to a complete stop. Assume that air resistance is negligible.

Calculate the distance between where the bus began to accelerate and the traffic lights where the bus stops.

Mark Scheme (4 marks):

List the known quantities:

- Initial acceleration, a_1 = 5.0 m s^{-2}
- Time for acceleration, t = 2.5 s
- Deceleration, a_2 = – 4.0 m s^{-2}
- Initial velocity, u = 3.0 m s^{-1}

Calculate the distance travelled by the bus whilst accelerating, S_1 :

- $s = S_1$ = ?, u = 3.0 m s^{-1}, v = not given, $a = a_1$ = 5.0 m s^{-2}, t = 2.5 s
- From the data booklet: $s = ut + \frac{1}{2}at^2$
- S_1 = (3.0 × 2.5) + ($\frac{1}{2}$× 5.0 × $(2.5)^2$) = 23.125 m *[1 mark]*

Calculate final velocity after acceleration, v:

- $s = S_1$, u = 3.0 ms^{-1}, v = ?, $a = a_1$ = 5.0 m s^{-2}, t = 2,5 s
- Recall: $v = u + at$
- v = 3.0 + (5.0 × 2.5) = 15.5 m s^{-1} *[1 mark]*

Calculate the distance travelled by the bus whilst decelerating, S_2:

- The bus reaches a maximum velocity of 15.5 m s^{-1} when accelerating. This is the new initial velocity for when it starts to decelerate
- $s = S_2$, u = 15.5 m s^{-1}, v = 0, $a = a_2$ = –4.0 m s^{-2}, t = not given
- From the data booklet: $v^2 = u^2 + 2as$
- $S_2 = \frac{v^2 - u^2}{2a_2}$
- $S_2 = \frac{0 - 15.5^2}{2 \times -4.0}$ = 30.0 m *[1 mark]*

Calculate the total distance travelled before stopping at the traffic lights, s:

- $s = S_1 + S_2$ = 23.125 + 30.0 = 53.125 = 53 m (2 s. f) *[1 mark]*

[Total: 4 marks]

*This question is split into two parts – when the bus is accelerating, and when it is decelerating. The velocity of the bus is increasing whilst it accelerates, therefore when we use the kinematics equation for deceleration, we need to make sure to have the correct **initial** velocity for this. This is equal to the **final** velocity **after** the **acceleration**. Remember that a deceleration is just a negative acceleration, so make sure a deceleration has a minus sign (e.g. –4 m s^{-2}) before substituting it into the kinematics equations!*

Using multiple kinematics equations is a common expectation in harder questions, so make sure to think through the modelling of the scenario before substituting any numbers!

Part (b) **[4 marks]**

On Fig. 1.1 draw the velocity–time graph for the motion of the bus.

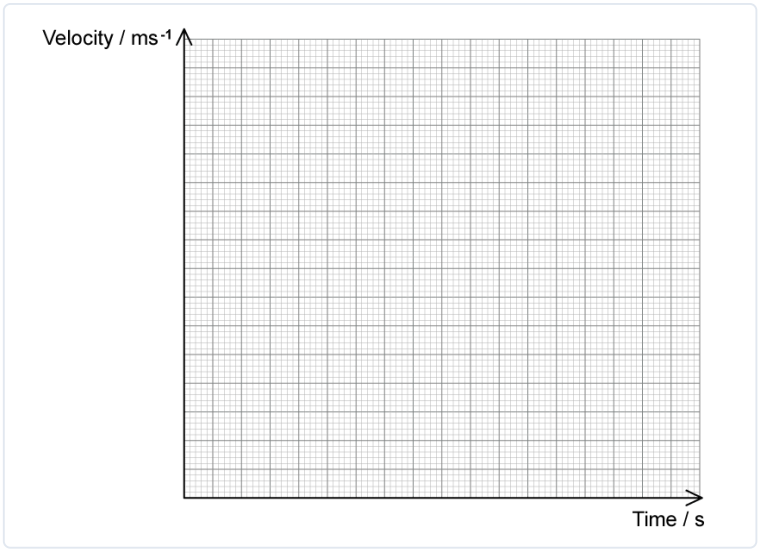


Fig. 1.1

Mark Scheme (4 marks):

List the known quantities:

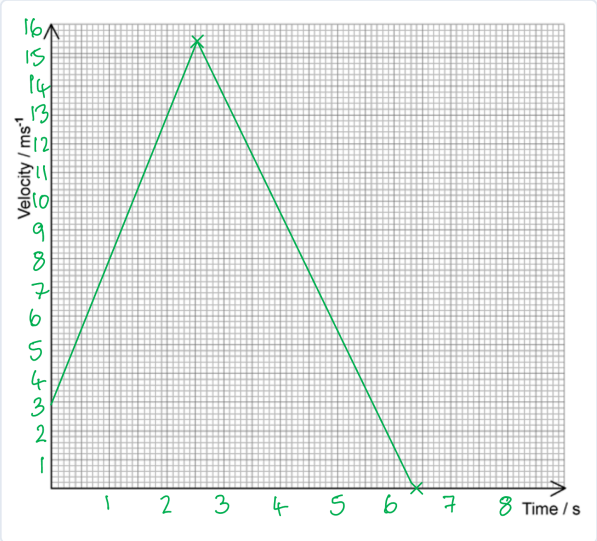
- Initial velocity = 3.0 m s^{-1}
- Final velocity after acceleration = 15.5 m s^{-1} (from part a)
- Deceleration = –4.0 m s^{-2}
- Time for acceleration, t_1 = 2.5 s
- Final velocity after deceleration = 0 m s^{-1} (at traffic lights)

Calculate the time taken to decelerate t_2:

- $s = S_2$ = not given in question, u = 15.5 m s^{-1}, v = 0, a = –4.0 m s^{-2}, $t = t_2$ = ?
- Recall: $v = u + at$
- $0 = 15.5 – 4.0t_2$

- $t_2 = \frac{15.5}{4.0}$ = 3.875 = 3.9 s *[1 mark]*

A graph that would score 3 marks should look like:



- Appropriate scale on time and velocity axis; *[1 mark]*
- Straight diagonal line upwards from 3 m s^{-1} to 15.5 m s^{-1} in 2.5 s; *[1 mark]*
- Straight diagonal line downwards from 15.5 m s^{-1} to 0 m s^{-1} at 6.4 s (2.5 + 3.9); *[1 mark]*

[Total: 4 marks]

When not given the axes on a table, think about the minimum and maximum values you need to plot (in this case 0 – 15.5 m s^{-1} and 0 – 6.4 s) and then pick an appropriate scale. This should be in either multiples of 1, 2, 5 or 10 for each large square and your graph must take up at least half the scales on both axes for full marks.

It can be helpful to mark the coordinates you know with an 'x' and then add in your straight lines with a ruler.

Error carried forward marks will apply here from part (a), although you will not get full marks in part (a). Therefore, make sure the shape of your graph is accurate and axes are all correctly labelled as per your value, and don't worry too much about getting the correct values if you are running out of time. This will help you get as many marks as possible.